Supporting Information for:

A Cobalt-Hydride Catalyst for the Hydrogenation of CO_2: Pathways for Catalysis and Deactivation

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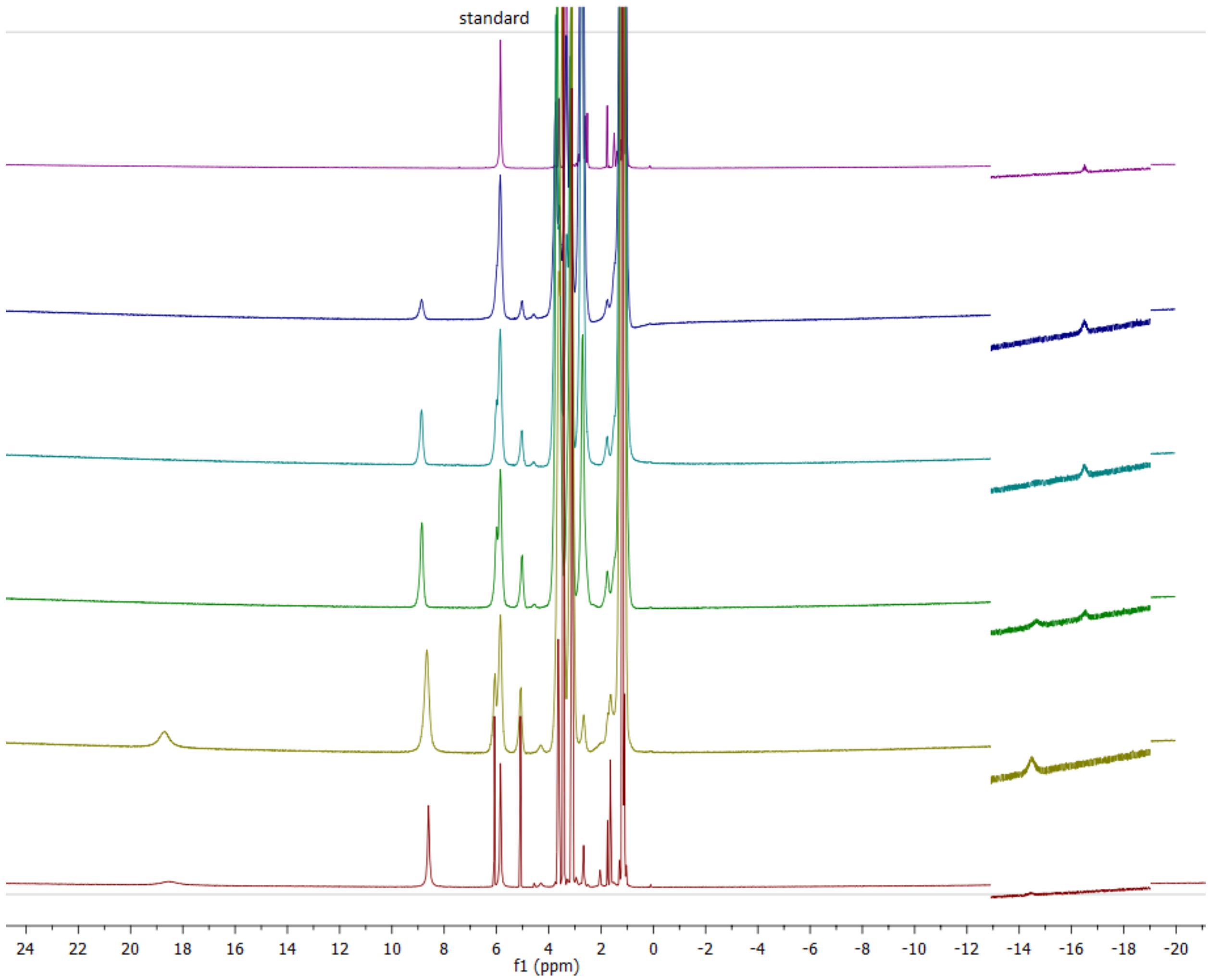

Figure S1. Representative time resolved ^1H NMR spectra for CO_2 hydrogenation with Co(dmpe)$_2$H. Catalytic conditions; 9.72 × 10^{-6} mol catalyst, 690 mM Vkd, 500 μL THF-d_8, 1.8 atm CO_2:H_2 (1:1 ratio), 21 °C. The region between -13 and -19 ppm is on a different intensity scale than the rest of the spectra for clarity. Time zero at the top. Note: This is not an experiment used for kinetic analysis as monitoring the hydrides would be too difficult considering the number of scans required at lower catalyst loadings compared to the rate at which catalysis occurs. This reaction is also mass transport limited enough to essentially allow for pseudo stop-flow kinetic methods to be used. Key species: HCOOH, 19 ppm; $HCOO^-$, -8.8 ppm; $VkdH^+$, 5/6 ppm (*d*); Co(dmpe)$_2$H$_2^+$, -14.5 ppm; Co(dmpe)$_2$H, -16.5 ppm.

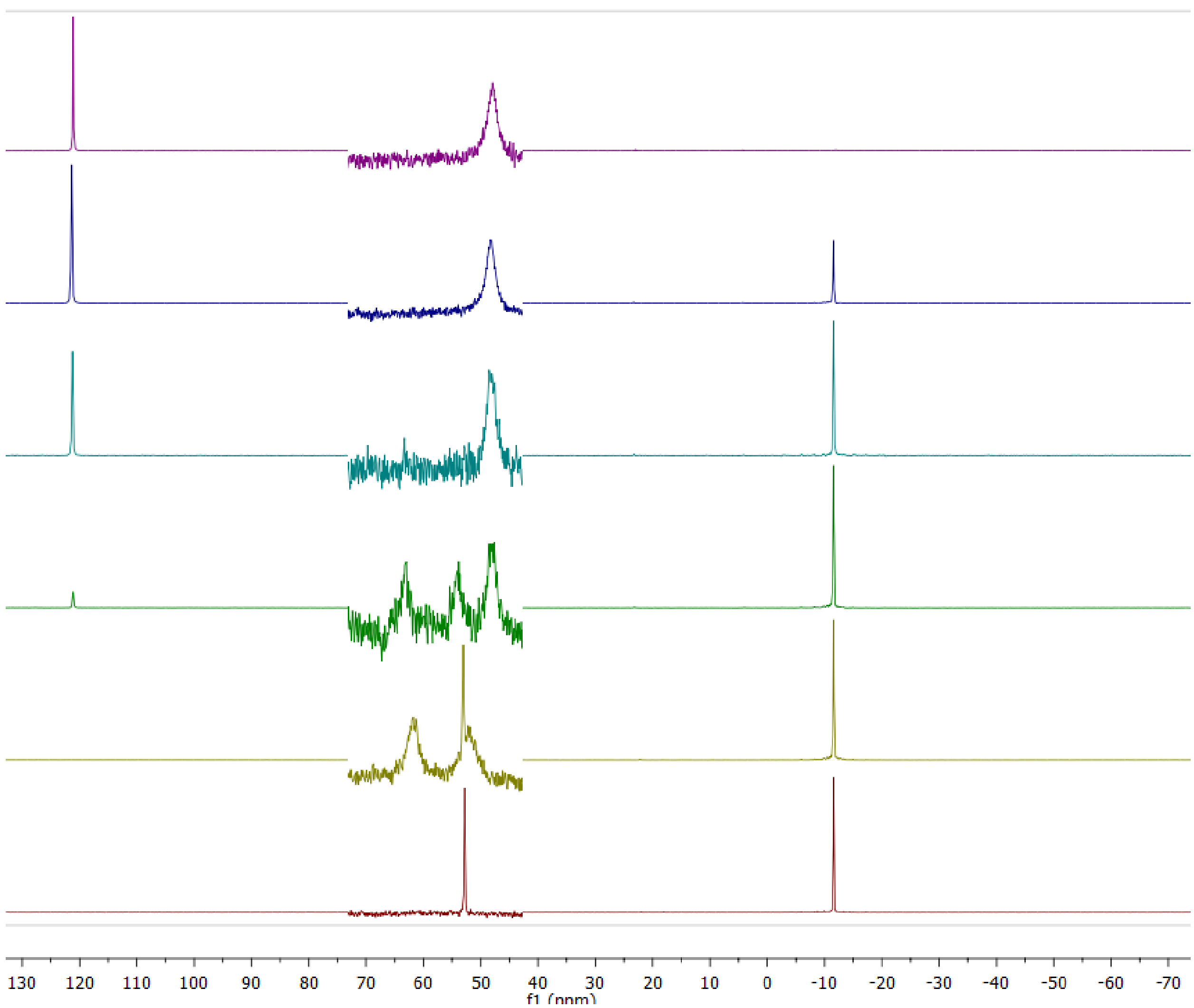

Figure S2. Representative time resolved $^{31}P\{^1H\}$ NMR spectra for CO_2 hydrogenation with $Co(dmpe)_2H$. Catalytic conditions; 9.72×10^{-6} mol catalyst, 690 mM Vkd, 500 μL THF-d_8, 1.8 atm CO_2:H_2 (1:1 ratio), 21 °C. The region between 45 and 75 ppm is on a different intensity scale than the rest of the spectra for clarity. Time zero at the top. Note: This is not an experiment used for kinetic analysis as monitoring the hydrides would be too difficult considering the number of scans required at lower catalyst loadings compared to the rate at which catalysis occurs. This reaction is also mass transport limited enough to essentially allow for pseudo stop-flow kinetic methods to be used. Key species: Vkd, 121 ppm; $Co(dmpe)_2H_2^+$, 52/61 ppm (*d*); $Co(dmpe)_2CO$, 53 ppm; $Co(dmpe)_2H$, 48 ppm; $VkdH^+$, -13 ppm.

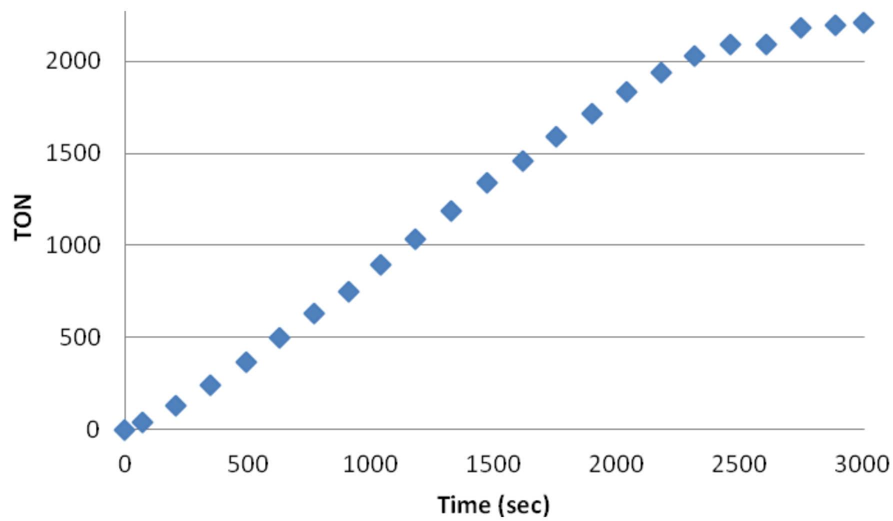

Figure S3. CO_2 hydrogenation kinetics using Co(dmpe)$_2$H as a function of TON. Catalytic conditions; 1.39 × 10^{-7} mol catalyst, 570 mM Vkd, 500 μL THF-d_8, 1 atm CO_2:H_2 (1:1 ratio), 21 °C (Table 1, entry 4).

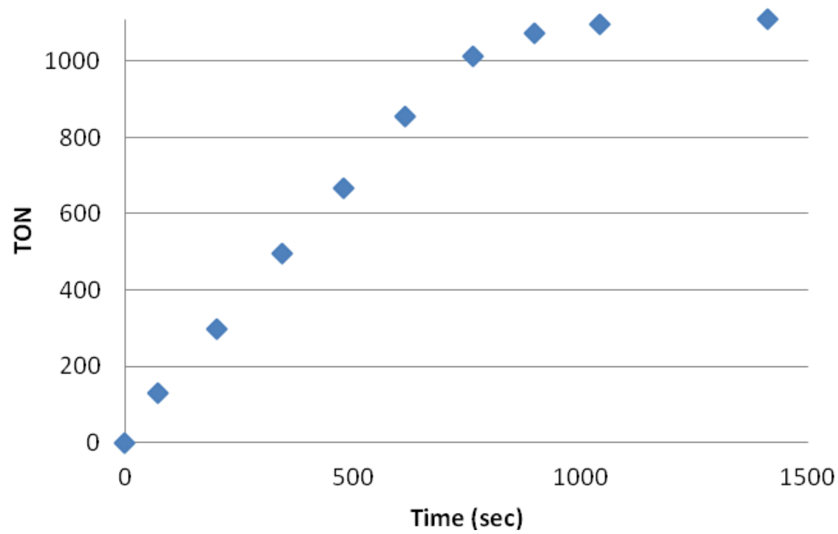

Figure S4. CO_2 hydrogenation kinetics using Co(dmpe)$_2$H as a function of TON. Catalytic conditions; 1.39 × 10^{-7} mol catalyst, 300 mM Vkd, 500 μL THF-d_8, 1.8 atm CO_2:H_2 (1:1 ratio), 21 °C (Table 1 entry 2).

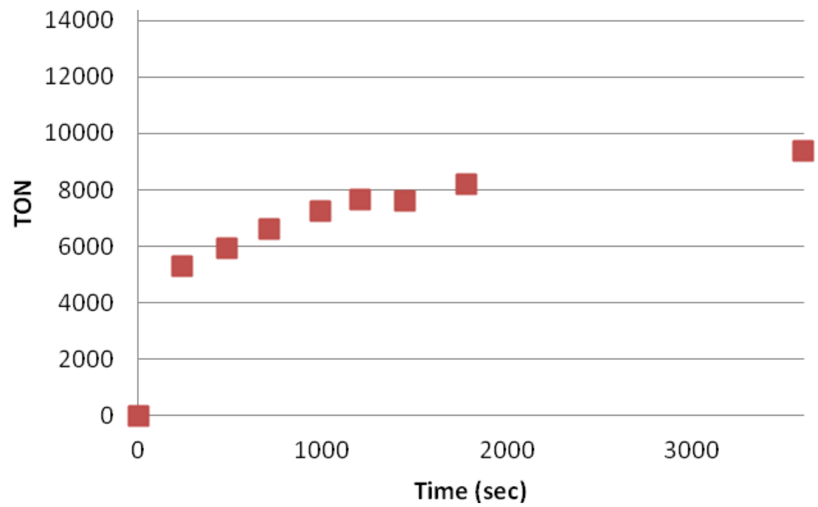

Figure S5. CO_2 hydrogenation kinetics using $Co(dmpe)_2H$ as a function of TON. Catalytic conditions; 2.0 × 10^{-8} mol catalyst, 510 mM Vkd, 500 μL THF-d_8, 20 atm CO_2:H_2 (1:1 ratio), 21 °C (Table 1, entry 5).

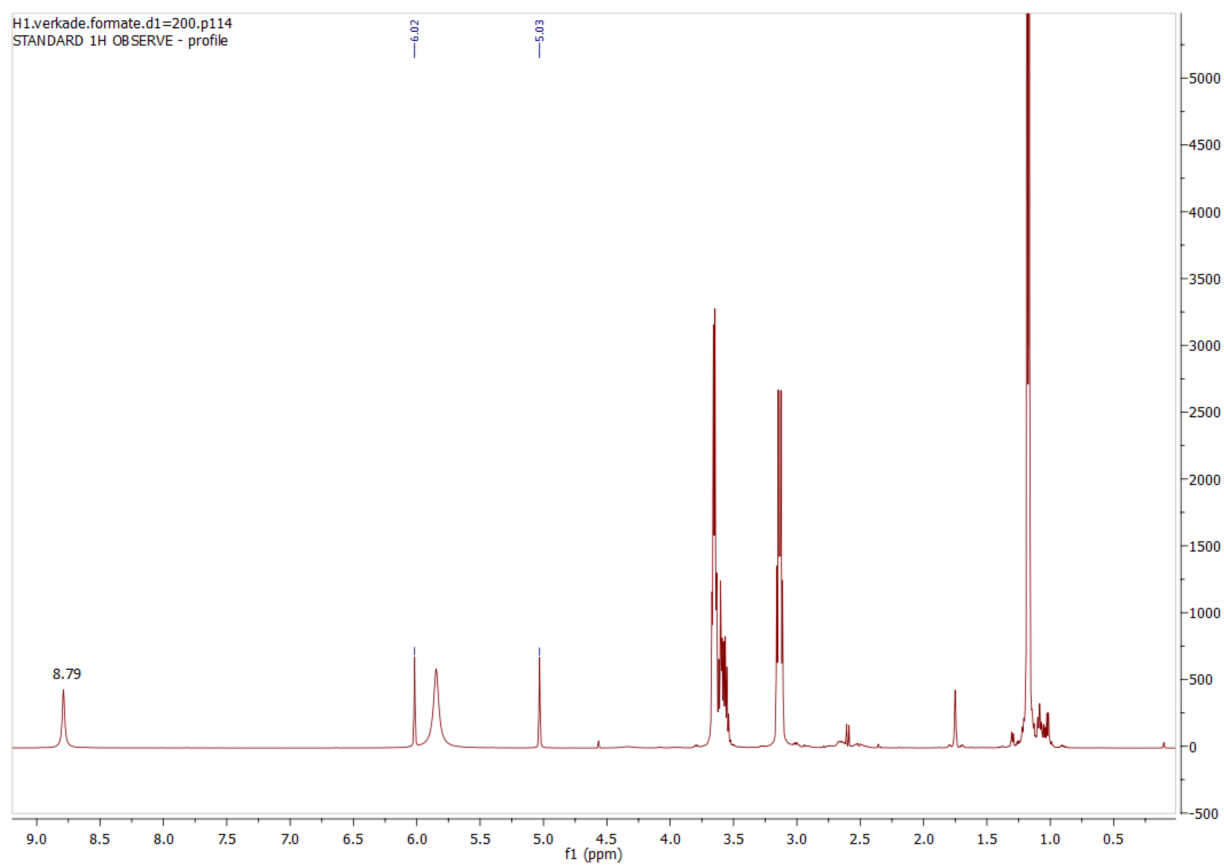

Figure S6. ^{1}H NMR spectrum of $[VkdH]^+[HCO_2]^-$ in THF-d_8 (separated from a catalytic reaction).

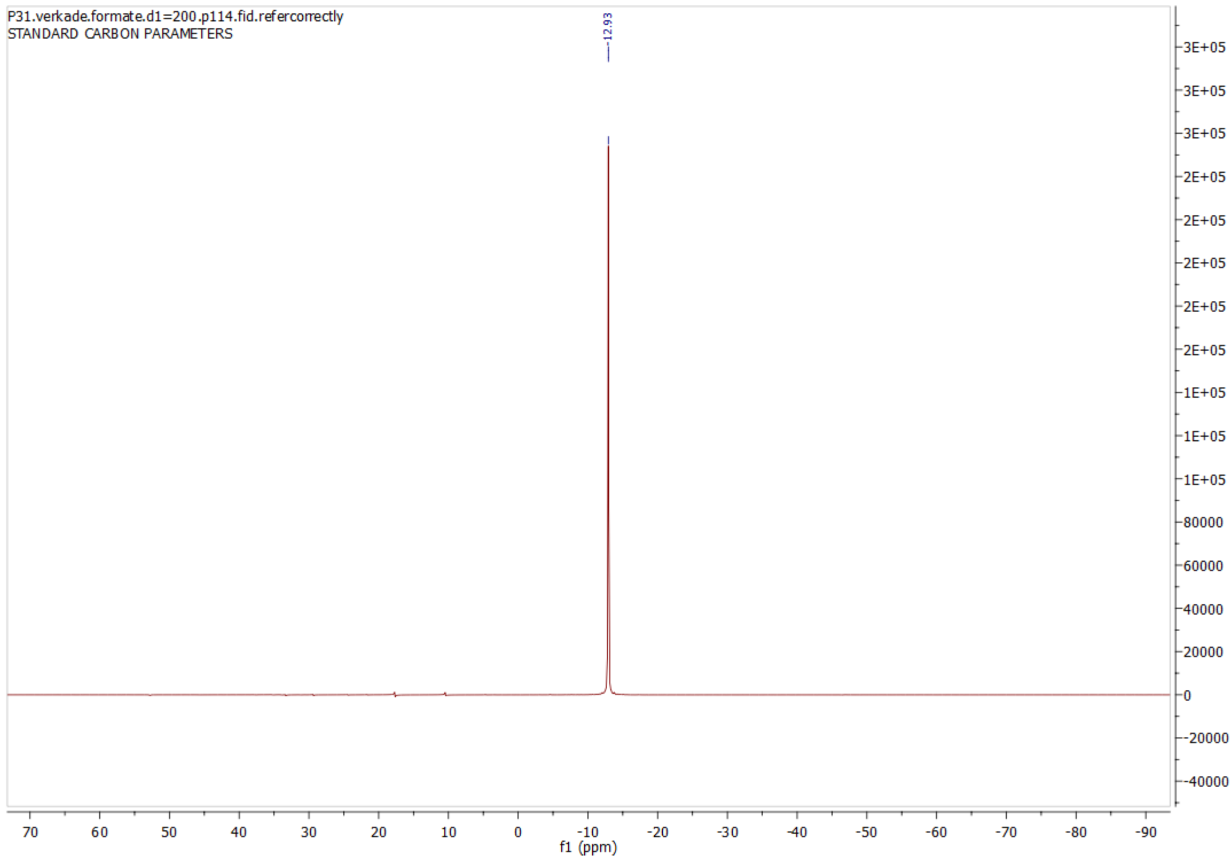

Figure S7. $^{31}P\{^{1}H\}$ NMR spectrum for $[VkdH]^{+}[HCO_2]^{-}$ in THF-d_8 referenced to H_3PO_4 in D_2O (separated from a catalytic reaction).

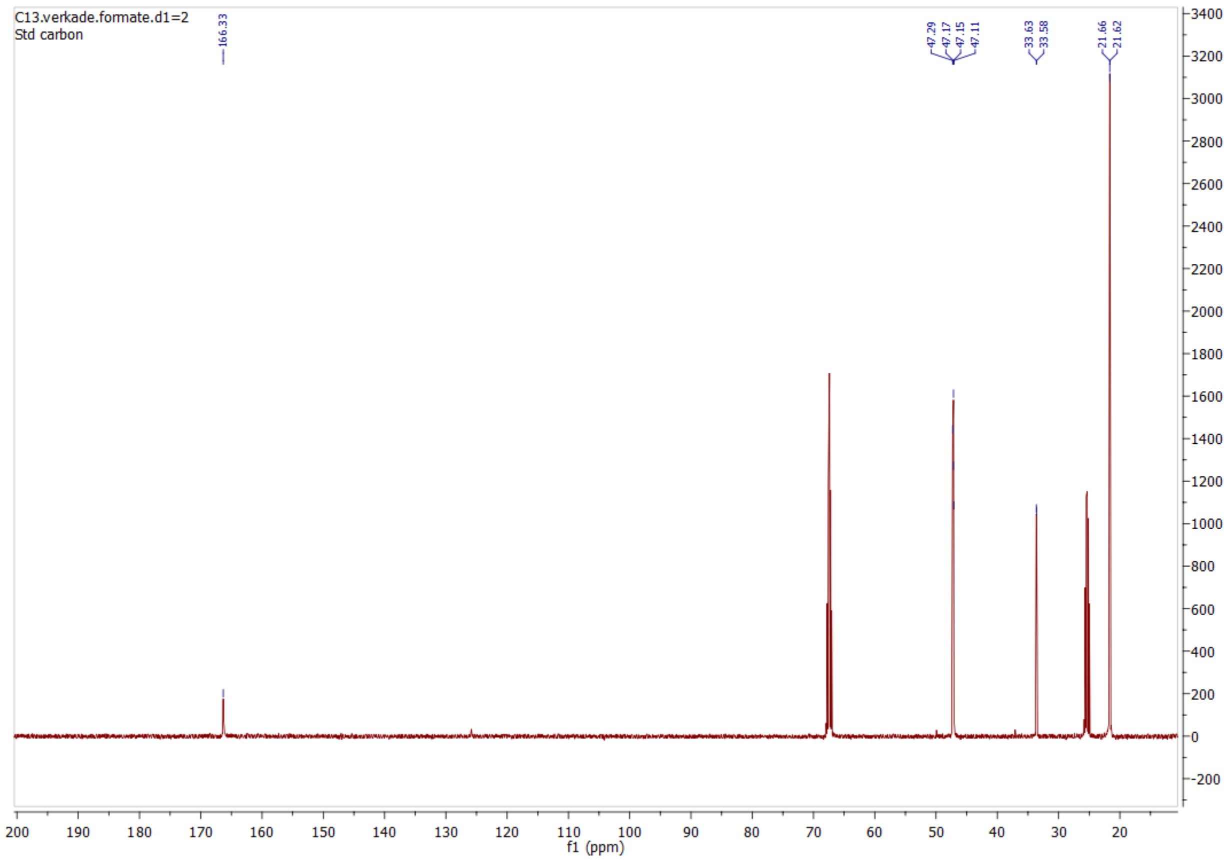

Figure S8. $^{13}C\ \{^{1}H\}$ NMR spectrum of $[VkdH]^{+}[HCO_2]^{-}$ in THF-d_8 (200 s delay time) (separated from a catalytic reaction).

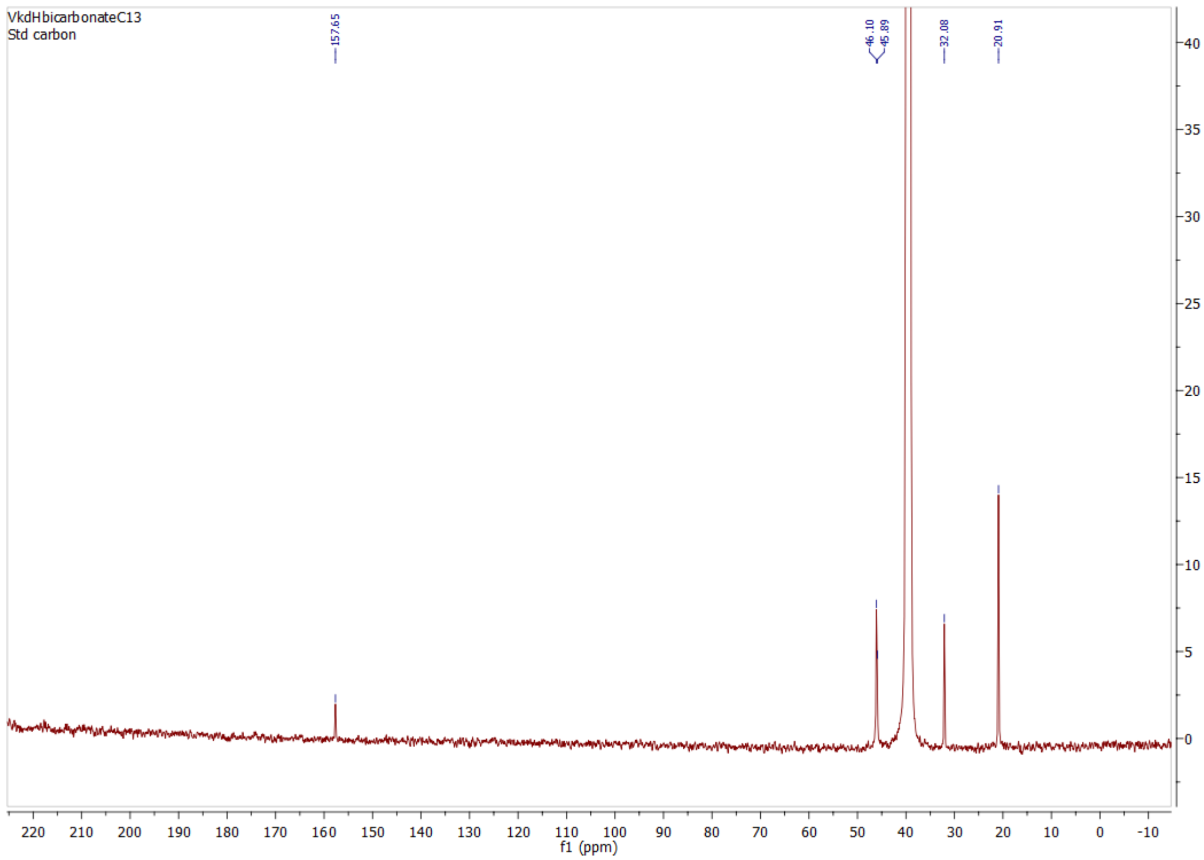

Figure S9. $^{13}C\ \{^{1}H\}$ NMR spectrum of $[VkdH]^{+}[HCO_3]^{-}$ in DMSO-d_6 (200 s delay time).

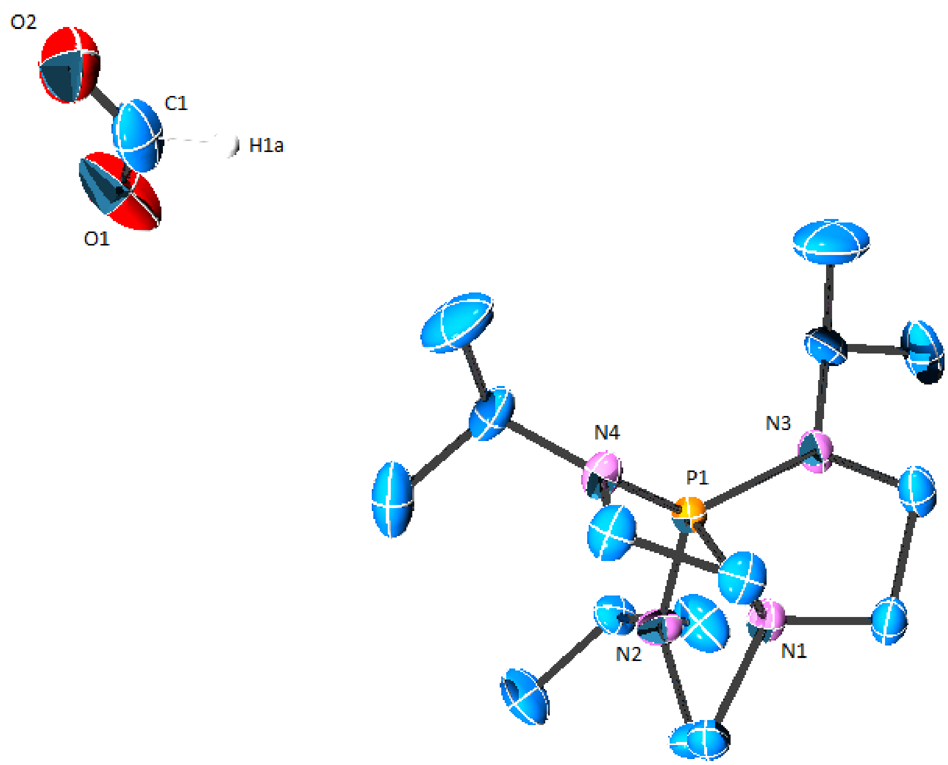

Figure S10. X-ray structure of $[VkdH]^+[HCO_2]^-$. Disorder on axial nitrogen (N1) removed for clarity.

Table S1. Crystal data and structure refinement for $[VkdH]^+[HCO_2]^-$.

Identification code	(VerkadeH)(formate)
Empirical formula	$C_{16}H_{35}N_4O_2P$
Formula weight	346.45
Temperature/K	145
Crystal system	triclinic
Space group	P-1
a/Å	9.4920(8)
b/Å	10.0652(9)
c/Å	11.6878(10)
α/°	72.135(2)
β/°	84.493(3)
γ/°	64.509(2)
Volume/$Å^3$	958.49(14)
Z	2
ρ_{calc} mg/mm^3	1.200
m/mm^{-1}	0.158
F(000)	380.0
Crystal size/mm^3	0.15 × 0.1 × 0.1
2Θ range for data collection	4.7 to 63.04°
Index ranges	-13 ≤ h ≤ 13, -14 ≤ k ≤ 14, -16 ≤ l ≤ 17
Reflections collected	22542
Independent reflections	6295[*R*(int) = 0.0303]
Data/restraints/parameters	6295/0/250
Goodness-of-fit on F^2	1.021
Final *R* indexes [I>=2σ (I)]	R_1 = 0.0448, wR_2 = 0.1109
Final *R* indexes [all data]	R_1 = 0.0650, wR_2 = 0.1222
Largest diff. peak/hole / e $Å^{-3}$	0.46/-0.31

Table S2. Fractional Atomic Coordinates (×10^4) and Equivalent Isotropic Displacement Parameters (Å^2×10^3) for $[VkdH]^+[HCO_2]^-$. U_{eq} is defined as 1/3 of the trace of the orthogonalized U_{IJ} tensor.

Atom	***x***	***y***	***z***	**U(eq)**
P1	8386.4(3)	6156.0(3)	2626.4(3)	18.67(8)
N1	7071.0(12)	7007.7(12)	1158.8(9)	22.8(2)
N2	9347.9(13)	7189.2(13)	1890.7(10)	25.7(2)
N3	8867.0(13)	4419.7(11)	2459.9(10)	25.3(2)
N4	6749.0(12)	6962.9(13)	3316.6(10)	24.7(2)
C2	10811.0(16)	6997.1(15)	2387.4(13)	27.6(3)
C5	9737.6(16)	2992.7(14)	3407.5(12)	28.0(3)
C13	8165.4(17)	4293.4(15)	1461.3(13)	29.7(3)
C11	8875.8(17)	8165.8(16)	652.9(12)	31.1(3)
C15	5263.4(15)	7889.8(17)	2622.4(13)	30.8(3)
C8	6816.2(17)	7187.7(18)	4494.8(13)	34.3(3)
C6	11008(2)	1809.9(17)	2888.4(17)	46.4(4)
C3	12235.8(18)	5945.8(19)	1873.7(18)	46.0(4)
C4	10840(2)	8551.4(19)	2203.6(18)	47.7(4)
C1	4101(2)	8052(2)	8417.5(18)	48.1(4)
C10	6848(2)	8731(2)	4355.4(18)	50.2(5)
C14B	6661(3)	5720(3)	1053(3)	27.6(6)
C7	8646(3)	2360(3)	4176(2)	70.7(7)
C12B	7992(3)	7412(3)	146(2)	27.0(6)
C9	5491(2)	7014(3)	5268.2(17)	62.7(6)
C16B	5638(3)	8323(3)	1301(3)	29.2(6)
C12A	7188(3)	8525(3)	486(2)	25.9(5)
C14A	7734(3)	5886(3)	489(2)	27.4(6)
C16A	5478(3)	7301(3)	1542(2)	29.3(6)
O1	5237.2(18)	8352.7(15)	8318.5(16)	71.4(5)
O2	3069(2)	8235.8(18)	9137.1(15)	80.5(5)

Table S3. Anisotropic Displacement Parameters ($Å^2×10^3$) for $[VkdH]^+[HCO_2]^-$. The Anisotropic displacement factor exponent takes the form: $-2\pi2[h2a*2U_{11}+...+2hka×b×U_{12}]$

Atom	U11	U22	U33	U23	U13	U12
P1	17.24(15)	18.71(14)	19.92(15)	-6.40(11)	-0.11(10)	-6.84(11)
N1	20.7(5)	23.4(5)	22.8(5)	-8.0(4)	-2.2(4)	-6.6(4)
N2	24.0(5)	30.1(5)	23.4(5)	-2.5(4)	-1.0(4)	-15.2(4)
N3	28.8(6)	19.1(4)	26.4(5)	-7.9(4)	-6.0(4)	-6.4(4)
N4	18.8(5)	30.7(5)	23.9(5)	-12.2(4)	1.5(4)	-7.2(4)
C2	27.4(6)	28.8(6)	32.1(7)	-7.0(5)	-1.9(5)	-17.5(5)
C5	33.4(7)	20.9(5)	29.2(7)	-2.1(5)	-5.0(5)	-13.4(5)
C13	31.4(7)	25.8(6)	33.6(7)	-12.7(5)	-6.6(5)	-9.3(5)
C11	34.1(7)	33.1(7)	23.5(6)	-0.1(5)	0.2(5)	-17.3(6)
C15	18.1(6)	38.1(7)	33.8(7)	-15.6(6)	0.0(5)	-6.0(5)
C8	28.7(7)	40.7(7)	24.2(7)	-14.3(6)	0.1(5)	-2.9(6)
C6	46.0(9)	24.9(7)	61.8(11)	-19.2(7)	-17.7(8)	-0.7(6)
C3	25.4(7)	38.3(8)	73.1(12)	-17.7(8)	3.5(7)	-11.7(6)
C4	53.7(10)	37.6(8)	66.5(12)	-19.4(8)	1.7(9)	-29.7(8)
C1	53.0(11)	41.5(9)	50.8(10)	-20.9(8)	-19.1(9)	-11.6(8)
C10	42.4(9)	53.7(10)	60.2(11)	-38.7(9)	-6.6(8)	-8.8(8)
C14B	24.4(13)	28.7(12)	32.6(14)	-13.0(11)	-6.2(11)	-9.7(10)
C7	96.4(18)	82.7(15)	50.2(12)	0.8(11)	11.7(11)	-69.1(15)
C12B	31.1(14)	28.7(12)	20.7(12)	-6.4(10)	0.5(10)	-12.7(11)
C9	58.6(12)	74.3(14)	36.2(10)	-13.2(9)	21.5(9)	-17.1(10)
C16B	22.5(13)	24.9(12)	33.0(15)	-11.7(11)	-7.0(11)	0.4(10)
C12A	28.5(13)	22.1(11)	24.0(12)	-4.3(9)	-3.0(9)	-8.6(9)
C14A	30.9(13)	26.3(11)	25.0(12)	-12(1)	-2.9(10)	-8.3(10)
C16A	19.5(12)	36.5(14)	33.2(14)	-12.0(11)	-2.7(10)	-11.2(10)
O1	69.9(10)	43.5(7)	100.8(12)	-2.0(7)	-50.5(9)	-26.1(7)
O2	108.1(14)	52.6(9)	65.5(10)	-27.1(8)	18.8(10)	-16.6(9)

Table S4. Bond Lengths for $[VkdH]^+[HCO_2]^-$.

AtomAtom	Length/Å	AtomAtom	Length/Å
P1 N1	1.9555(11)	N4 C8	1.4737(17)
P1 N2	1.6647(11)	C2 C3	1.520(2)
P1 N3	1.6750(10)	C2 C4	1.5246(19)
P1 N4	1.6736(11)	C5 C6	1.519(2)
N1 C14B	1.546(3)	C5 C7	1.525(2)
N1 C12B	1.464(3)	C13 C14B	1.508(3)
N1 C16B	1.474(3)	C13 C14A	1.561(3)
N1 C12A	1.535(3)	C11 C12B	1.597(3)
N1 C14A	1.456(3)	C11 C12A	1.501(3)
N1 C16A	1.463(3)	C15 C16B	1.522(3)
N2 C2	1.4689(16)	C15 C16A	1.514(3)
N2 C11	1.4604(17)	C8 C10	1.525(2)
N3 C5	1.4665(16)	C8 C9	1.518(2)
N3 C13	1.4611(16)	C1 O1	1.226(2)
N4 C15	1.4629(16)	C1 O2	1.218(3)

Table S5. Bond Angles for $[VkdH]^{+}[HCO_2]^{-}$.

AtomAtomAtom	Angle/°	AtomAtomAtom	Angle/°
N2 P1 N1	86.76(5)	C13 N3 P1	119.79(9)
N2 P1 N3	119.47(6)	C13 N3 C5	117.40(10)
N2 P1 N4	120.22(6)	C15 N4 P1	119.85(9)
N3 P1 N1	86.67(5)	C15 N4 C8	116.49(10)
N4 P1 N1	86.72(5)	C8 N4 P1	120.85(9)
N4 P1 N3	119.34(6)	N2 C2 C3	111.86(12)
C14B N1 P1	106.71(12)	N2 C2 C4	111.16(12)
C12B N1 P1	107.58(12)	C3 C2 C4	111.22(13)
C12B N1 C14B	110.40(17)	N3 C5 C6	111.29(12)
C12B N1 C16B	114.01(18)	N3 C5 C7	111.41(14)
C12B N1 C12A	47.68(15)	C6 C5 C7	112.36(15)
C16B N1 P1	107.90(12)	N3 C13 C14B	108.18(14)
C16B N1 C14B	109.93(17)	N3 C13 C14A	103.88(13)
C16B N1 C12A	70.02(16)	C14B C13 C14A	46.17(15)
C12A N1 P1	105.58(11)	N2 C11 C12B	104.38(13)
C12A N1 C14B	145.72(16)	N2 C11 C12A	106.43(14)
C14A N1 P1	106.46(12)	C12A C11 C12B	46.00(14)
C14A N1 C14B	47.20(16)	N4 C15 C16B	107.45(14)
C14A N1 C12B	65.69(16)	N4 C15 C16A	104.83(13)
C14A N1 C16B	143.49(17)	C16A C15 C16B	40.61(15)
C14A N1 C12A	111.78(16)	N4 C8 C10	111.47(13)
C14A N1 C16A	115.17(17)	N4 C8 C9	111.68(14)
C16A N1 P1	106.52(12)	C9 C8 C10	111.22(15)
C16A N1 C14B	70.44(16)	O2 C1 O1	130.50(19)
C16A N1 C12B	143.75(17)	C13 C14B N1	102.87(17)
C16A N1 C16B	42.04(16)	N1 C12B C11	103.01(17)
C16A N1 C12A	110.64(16)	N1 C16B C15	105.86(18)
C2 N2 P1	121.71(9)	C11 C12A N1	104.26(16)
C11 N2 P1	120.45(9)	N1 C14A C13	104.64(17)
C11 N2 C2	117.32(11)	N1 C16A C15	106.80(17)
C5 N3 P1	121.76(8)		

Table S6. Hydrogen Atom Coordinates (Å×10^4) and Isotropic Displacement Parameters (Å^2×10^3) for $[VkdH]^+[HCO_2]^-$.

Atom	*x*	*y*	*z*	U(eq)
H2	10855	6495	3274	33
H5	10268	3256	3947	34
H13A	7954	3365	1725	36
H13B	8887	4202	790	36
H13C	7223	4109	1711	36
H13D	8918	3446	1155	36
H11A	9798	8163	178	37
H11B	8170	9239	630	37
H11C	9517	7620	80	37
H11D	9007	9126	516	37
H15A	4661	8829	2874	37
H15B	4633	7291	2756	37
H15C	5040	8997	2367	37
H15D	4394	7748	3104	37
H8	7817	6360	4922	41
H6A	10532	1543	2336	70
H6B	11589	882	3543	70
H6C	11723	2241	2452	70
H3A	12164	4974	1972	69
H3B	13183	5736	2301	69
H3C	12279	6449	1017	69
H4A	11008	8982	1356	72
H4B	11690	8415	2706	72
H4C	9842	9256	2430	72
H10A	7785	8761	3938	75
H10B	6864	8858	5152	75
H10C	5916	9567	3886	75
H14A	5801	5644	1580	33
H14B	6357	5900	213	33
H7A	7857	3155	4499	106
H7B	9250	1460	4841	106
H7C	8129	2056	3681	106
H12A	7309	8158	-568	32
H12B	8743	6486	-71	32
H9A	4495	7855	4902	94
H9B	5640	7051	6074	94
H9C	5484	6025	5327	94
H16A	4770	8502	782	35
H16B	5811	9271	1081	35
H12C	6890	8843	-378	31
H12D	6503	9357	834	31
H14C	8674	5946	75	33
H14D	6962	6062	-117	33
H16C	5327	6341	1762	35
H16D	4711	8078	885	35
H1	9265(15)	5578(14)	3613(11)	10(3)
H1A	4020(40)	7530(40)	7810(30)	114(10)

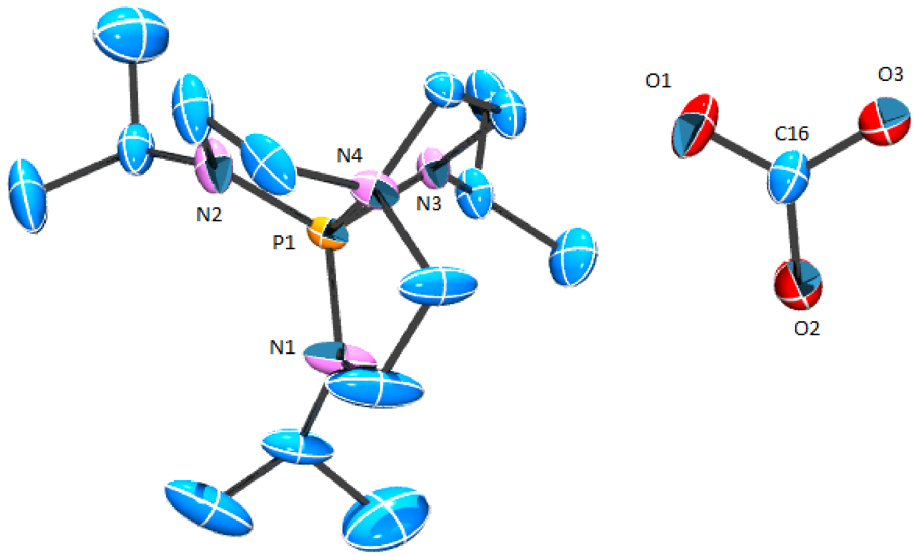

Figure S11. X-ray structure of $[VkdH]^{+}[HCO_3]^{-}$.

Table S7. Crystal data and structure refinement for $[VkdH]^+[HCO_3]^-$.

Identification code	$[VerkadesH][HCO_3]$
Empirical formula	$C_{16}H_{39}N_4O_5P$
Formula weight	398.48
Temperature/K	143.15
Crystal system	triclinic
Space group	P-1
a/Å	8.8946(19)
b/Å	9.599(2)
c/Å	14.595(3)
α/°	105.460(6)
β/°	94.446(6)
γ/°	114.780(6)
Volume/Å^3	1064.5(4)
Z	2
ρ_{calc} mg/mm^3	1.243
m/mm-1	0.161
F(000)	436.0
Crystal size/mm^3	0.2 × 0.045 × 0.02
2θ range for data collection	4.86 to 55.8°
Index ranges	$-11 \le h \le 11, -12 \le k \le 12, -19 \le l \le 19$
Reflections collected	18554
Independent reflections	5060[*R*(int) = 0.0769]
Data/restraints/parameters	5060/0/252
Goodness-of-fit on F^2	1.033
Final R indexes [I>=2σ (I)]	$R_1 = 0.0684$, $wR_2 = 0.1579$
Final R indexes [all data]	$R_1 = 0.1368$, $wR_2 = 0.1909$
Largest diff. peak/hole / e Å^{-3}	0.62/-0.58

Table S8. Fractional Atomic Coordinates (×10^4) and Equivalent Isotropic Displacement Parameters (Å^2×10^3) for $[VkdH]^+[HCO_3]^-$. U_{eq} is defined as 1/3 of the trace of the orthogonalized U_{IJ} tensor.

Atom	***x***	***y***	***z***	**U(eq)**
P1	1749.1(10)	6232.2(10)	2131.7(6)	22.3(2)
O1	2711(3)	3064(3)	4518(2)	48.7(7)
O2	52(3)	1356(3)	4465(2)	48.8(7)
O3	2213(3)	998(3)	5046(2)	41.7(7)
O4	5949(3)	3698(3)	4121.9(19)	43.0(7)
O5	1823(5)	7774(5)	6710(3)	76.6(10)
N1	15(4)	6072(4)	2526(2)	43.2(9)
N2	3327(4)	8068(3)	2291(2)	35.1(7)
N3	2070(3)	4599(3)	1760.7(18)	23.0(6)
N4	2736(3)	6501(3)	3441.0(18)	27.3(6)
C1	-38(6)	6195(8)	3534(3)	68.2(16)
C2	1302(5)	5755(6)	3884(3)	48.7(11)
C3	4594(6)	8988(4)	3211(3)	48.3(11)
C4	3771(5)	8289(4)	3949(3)	44(1)
C5	2840(4)	4140(4)	2474(2)	30.5(8)
C6	3811(4)	5654(4)	3332(2)	24.6(7)
C7	-1372(5)	6153(6)	1966(3)	47.0(11)
C8	-1570(7)	7625(7)	2495(4)	79.1(18)
C9	-3010(6)	4584(8)	1712(4)	89.8(19)
C10	3625(5)	8759(4)	1498(3)	36.6(9)
C11	5256(6)	8883(7)	1195(4)	73.5(15)
C12	3518(7)	10353(5)	1756(3)	62.7(14)
C13	1074(5)	3310(4)	825(2)	31.9(8)
C14	-313(5)	1821(5)	966(3)	49.5(11)
C15	2238(5)	2883(5)	219(3)	42.5(10)
C16	1668(5)	1828(4)	4642(2)	32.9(8)

Table S9. Anisotropic Displacement Parameters ($Å^2×10^3$) for $[VkdH]^+[HCO_3]^-$. The Anisotropic displacement factor exponent takes the form: $-2π2[h2a*2U_{11}+...+2hka×b×U_{12}]$

Atom	**U_{11}**	**U_{22}**	**U_{33}**	**U_{23}**	**U_{13}**	**U_{12}**
P1	21.9(4)	25.5(5)	18.5(4)	1.8(3)	1.3(3)	13.7(4)
O1	43.9(16)	29.5(15)	55.7(18)	19.0(13)	6.3(14)	-0.8(13)
O2	35.1(15)	37.8(16)	75(2)	25.2(15)	4.8(14)	15.5(13)
O3	32.4(14)	40.0(16)	54.0(17)	20.3(13)	8.1(13)	15.0(13)
O4	46.1(16)	41.3(16)	40.6(16)	8.9(13)	6.7(13)	22.2(14)
O5	71(3)	77(3)	80(3)	26(2)	7(2)	34(2)
N1	33.9(17)	88(3)	22.3(15)	18.0(16)	9.7(13)	41.0(19)
N2	43.1(18)	20.7(15)	29.3(16)	1.0(12)	-11.8(13)	11.1(14)
N3	27.1(14)	15.9(13)	20.0(13)	1.4(11)	-2.0(11)	8.4(11)
N4	26.3(14)	35.7(16)	20.9(14)	4.9(12)	2.1(11)	18.3(13)
C1	49(3)	154(5)	35(2)	40(3)	22(2)	68(3)
C2	32(2)	100(4)	28(2)	30(2)	14.2(16)	36(2)
C3	63(3)	23(2)	40(2)	2.9(17)	-22(2)	12.4(19)
C4	57(3)	35(2)	28(2)	-13.6(16)	-16.2(18)	29(2)
C5	37.1(19)	21.6(17)	28.6(18)	6.1(14)	-1.6(15)	12.4(16)
C6	21.1(16)	26.3(17)	27.8(17)	9.2(14)	1.4(13)	12.5(14)
C7	33(2)	87(3)	32(2)	17(2)	6.7(17)	40(2)
C8	94(4)	137(5)	55(3)	29(3)	26(3)	98(4)
C9	44(3)	125(5)	73(4)	35(4)	-11(3)	17(3)
C10	38(2)	26.2(19)	39(2)	13.8(16)	-7.0(17)	10.0(17)
C11	64(3)	111(4)	68(3)	48(3)	19(3)	48(3)
C12	82(3)	30(2)	70(3)	15(2)	-14(3)	26(2)
C13	42(2)	21.8(17)	23.2(17)	0.2(14)	-4.0(15)	13.2(16)
C14	46(2)	27(2)	47(2)	0.6(18)	-6.9(19)	1.4(19)
C15	70(3)	34(2)	28.0(19)	3.7(16)	8.4(19)	32(2)
C16	34(2)	26.1(19)	29.6(19)	10.5(16)	2.2(16)	5.9(17)

Table S10. Bond Lengths for $[VkdH]^+[HCO_3]^-$.

AtomAtom	**Length/Å**	**AtomAtom**	**Length/Å**
P1 N1	1.650(3)	N4 C2	1.472(4)
P1 N2	1.669(3)	N4 C4	1.495(4)
P1 N3	1.666(3)	N4 C6	1.486(4)
P1 N4	1.943(3)	C1 C2	1.517(5)
O1 C16	1.232(4)	C3 C4	1.498(6)
O2 C16	1.293(4)	C5 C6	1.495(4)
O3 C16	1.321(4)	C7 C8	1.507(6)
N1 C1	1.450(5)	C7 C9	1.519(7)
N1 C7	1.467(4)	C10 C11	1.515(6)
N2 C3	1.467(4)	C10 C12	1.521(5)
N2 C10	1.469(4)	C13 C14	1.525(5)
N3 C5	1.461(4)	C13 C15	1.525(5)
N3 C13	1.474(4)		

Table S11. Bond Angles for $[VkdH]^{+}[HCO_3]^{-}$.

AtomAtomAtom	**Angle/°**	**AtomAtomAtom**	**Angle/°**
N1 P1 N2	119.77(17)	C6 N4 C4	111.6(3)
N1 P1 N3	119.79(16)	N1 C1 C2	105.6(3)
N1 P1 N4	87.17(13)	N4 C2 C1	105.5(3)
N2 P1 N4	86.58(13)	N2 C3 C4	105.5(3)
N3 P1 N2	119.52(14)	N4 C4 C3	106.5(3)
N3 P1 N4	86.70(12)	N3 C5 C6	106.1(3)
C1 N1 P1	120.0(2)	N4 C6 C5	105.5(2)
C1 N1 C7	116.2(3)	N1 C7 C8	110.8(3)
C7 N1 P1	122.6(2)	N1 C7 C9	111.2(4)
C3 N2 P1	120.6(2)	C8 C7 C9	112.5(4)
C3 N2 C10	116.8(3)	N2 C10 C11	111.5(3)
C10 N2 P1	122.3(2)	N2 C10 C12	111.0(3)
C5 N3 P1	119.6(2)	C11 C10 C12	113.7(4)
C5 N3 C13	116.6(2)	N3 C13 C14	112.0(3)
C13 N3 P1	120.2(2)	N3 C13 C15	110.6(3)
C2 N4 P1	106.2(2)	C15 C13 C14	112.3(3)
C2 N4 C4	112.9(3)	O1 C16 O2	123.3(3)
C2 N4 C6	112.6(3)	O1 C16 O3	119.2(3)
C4 N4 P1	106.5(2)	O2 C16 O3	117.3(3)
C6 N4 P1	106.40(19)		

Table S12. Hydrogen Atom Coordinates (Å×10^4) and Isotropic Displacement Parameters (Å^2×10^3) for $[VkdH]^+[HCO_3]^-$.

Atom	***x***	***y***	***z***	**U(eq)**
H1	1100(30)	6030(30)	1233(18)	4(6)
H3	1398	305	5189	63
H4C	4867	3269	4109	64
H4D	6439	4699	4512	64
H5C	1323	7946	6247	115
H5D	2522	7416	6493	115
H1A	-1173	5433	3585	82
H1B	218	7313	3927	82
H2A	1654	6190	4603	58
H2B	864	4560	3670	58
H3A	4899	10159	3396	58
H3B	5633	8859	3154	58
H4A	4641	8503	4505	53
H4B	3038	8780	4198	53
H5A	3608	3716	2200	37
H5B	1955	3291	2665	37
H6A	4008	5384	3925	30
H6B	4921	6345	3216	30
H7	-1054	6280	1340	56
H8A	-1937	7513	3099	119
H8B	-2420	7717	2080	119
H8C	-481	8602	2650	119
H9A	-2854	3680	1308	135
H9B	-3919	4680	1352	135
H9C	-3314	4380	2311	135
H10	2679	7977	924	44
H11A	6224	9679	1729	110
H11B	5354	9230	619	110
H11C	5243	7819	1040	110
H12A	2422	10180	1923	94
H12B	3629	10745	1198	94
H12C	4435	11162	2314	94
H13	501	3758	456	38
H14A	207	1300	1283	74
H14B	-1031	1050	332	74
H14C	-1006	2155	1376	74
H15A	3058	3859	109	64
H15B	1560	2053	-409	64
H15C	2845	2464	567	64

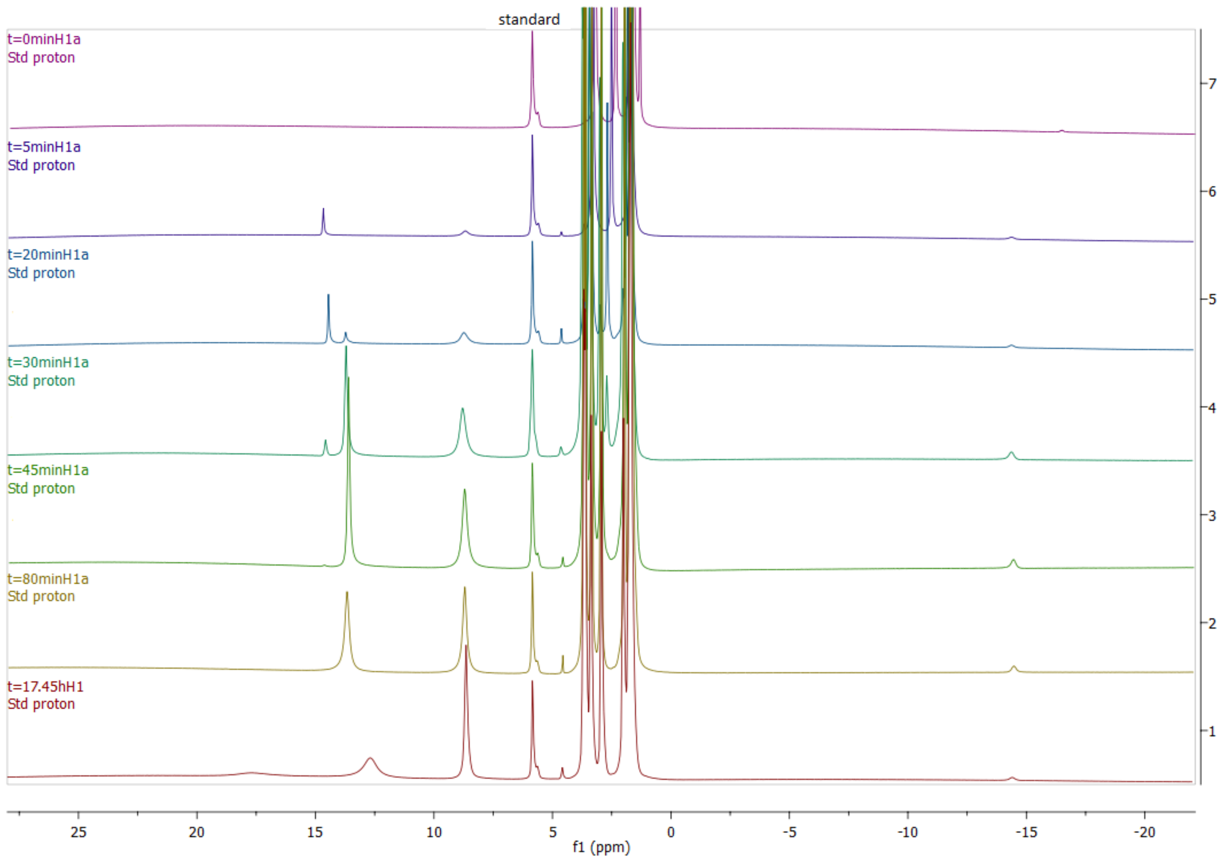

Figure S12. Representative time resolved ^{1}H NMR spectra for CO_2 hydrogenation with Co(dmpe)$_2$H (low concentration) and DBU. Catalytic conditions; 1.39×10^{-5} mol catalyst, 960 mM DBU, 300 μL THF-d_8, 40 atm CO_2:H_2 (1:1 ratio), 21 °C.

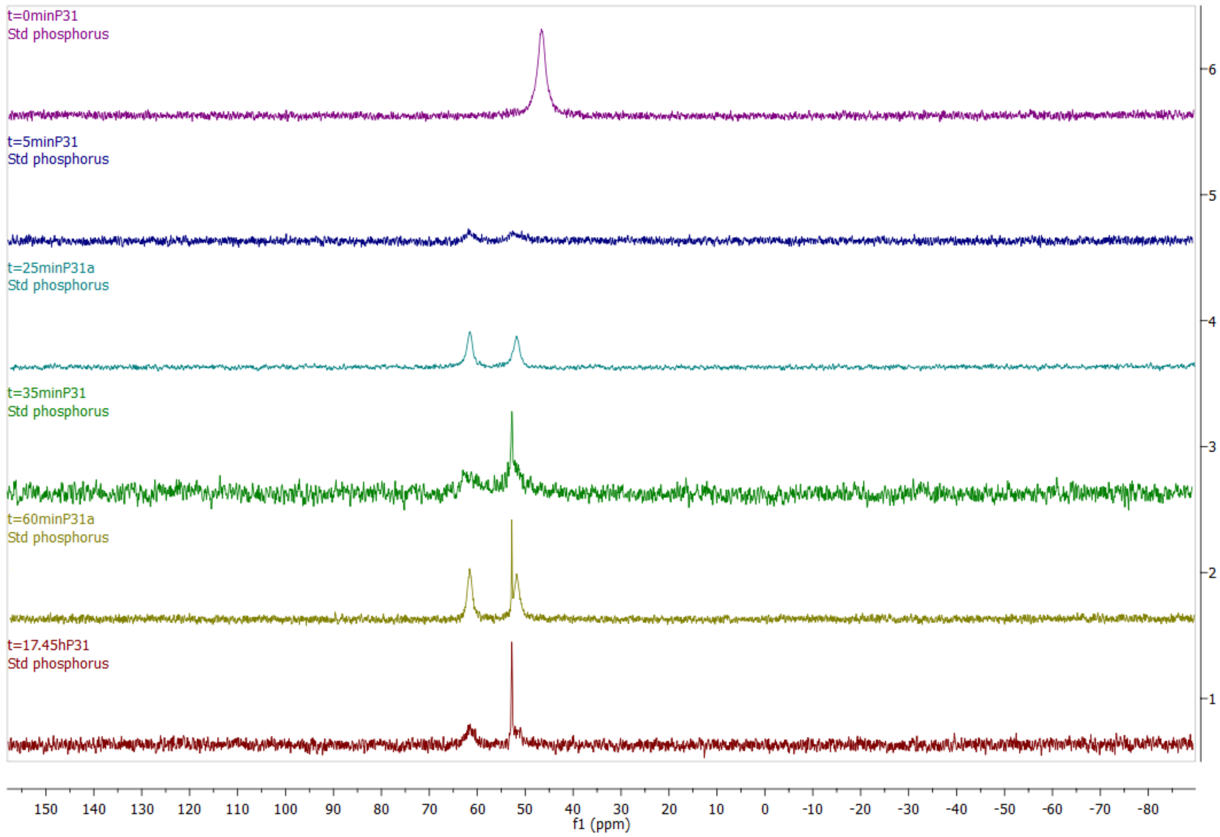

Figure S13. Representative time resolved ^{31}P NMR spectra for CO_2 hydrogenation with Co(dmpe)$_2$H (low concentration) and DBU. Catalytic conditions; 1.39 × 10^{-5} mol catalyst, 960 mM DBU, 300 μL THF-d_8, 40 atm CO_2:H_2 (1:1 ratio), 21 °C.

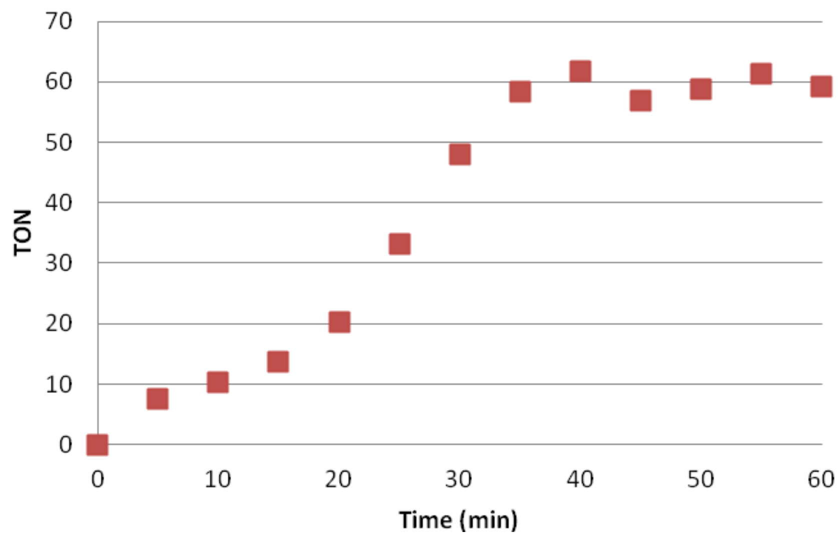

Figure S14. CO_2 hydrogenation kinetics using $Co(dmpe)_2H$ (P, 300 psi) as a function of TON. Catalytic conditions; 1.42×10^{-5} mol catalyst, 960 mM DBU, 300 μL THF-d_8, 20 atm CO_2:H_2 (1:1 ratio), 21 °C; assumes a formate to DBU ratio of 2:1 from literature precedent (table 3, entry 5).

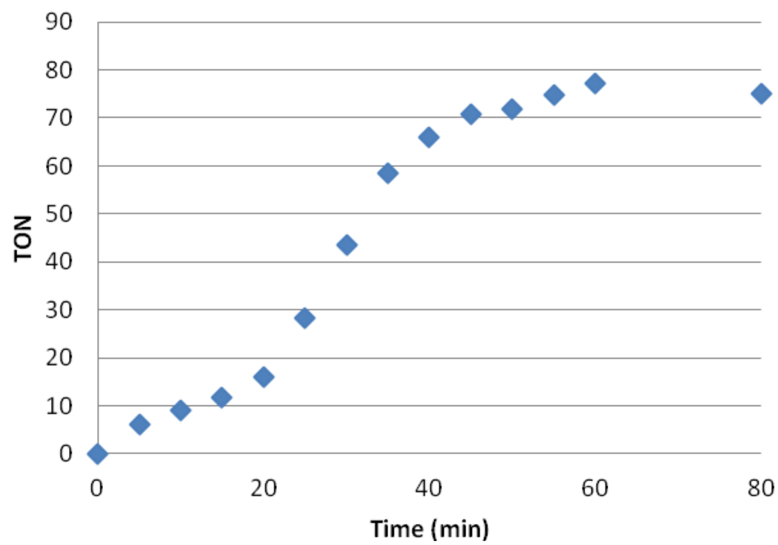

Figure S15. CO_2 hydrogenation kinetics using $[Co(dmpe)_2H_2]BF_4$ as a function of TON. Catalytic conditions; 1.12×10^{-5} mol catalyst, 960 mM DBU, 300 μL THF-d_8, 40 atm CO_2:H_2 (1:1 ratio), 21 °C; assumes a formate to DBU ratio of 2:1 from literature precedent (table 3, entry 8).

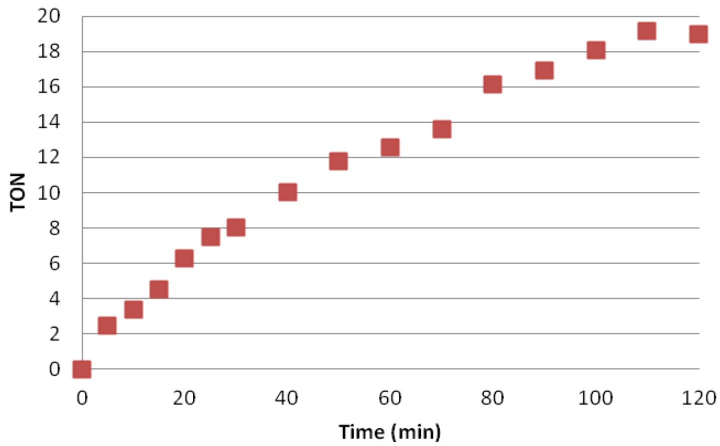

Figure S16. CO_2 hydrogenation kinetics using Rh(dmpe)$_2$H as a function of TON. Catalytic conditions; 1.43 × 10^{-5} mol catalyst, 960 mM DBU , 300 μL THF-d_8, 40 atm CO_2:H_2 (1:1 ratio), 21 °C; assumes a formate to DBU ratio of 2:1 from literature precedent (table 3 entry 9).

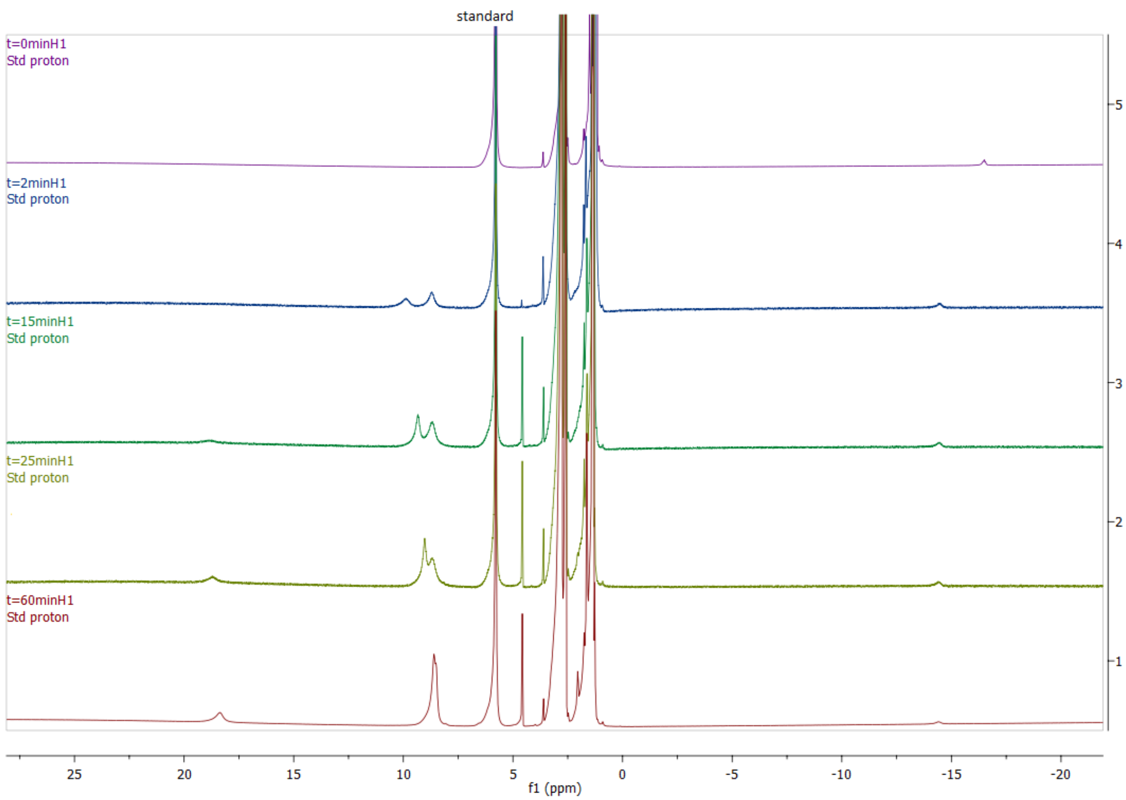

Figure S17. Representative time resolved ^{1}H NMR spectra for CO_2 hydrogenation with Co(dmpe)$_2$H. Catalytic conditions; 1.42 × 10^{-5} mol catalyst, 710 mM P$_1^t$Bu, 300 μL THF-d_8, 20 atm CO_2:H_2 (1:1 ratio), 21 °C. Key species: HCOOH, 19 ppm; $HCOO^-$, -8.8 ppm; P$_1^t$BuH$^+$, moves from 10 to 8.7 ppm; Co(dmpe)$_2$H$_2^+$, -14.5 ppm; Co(dmpe)$_2$H, -16.5 ppm.

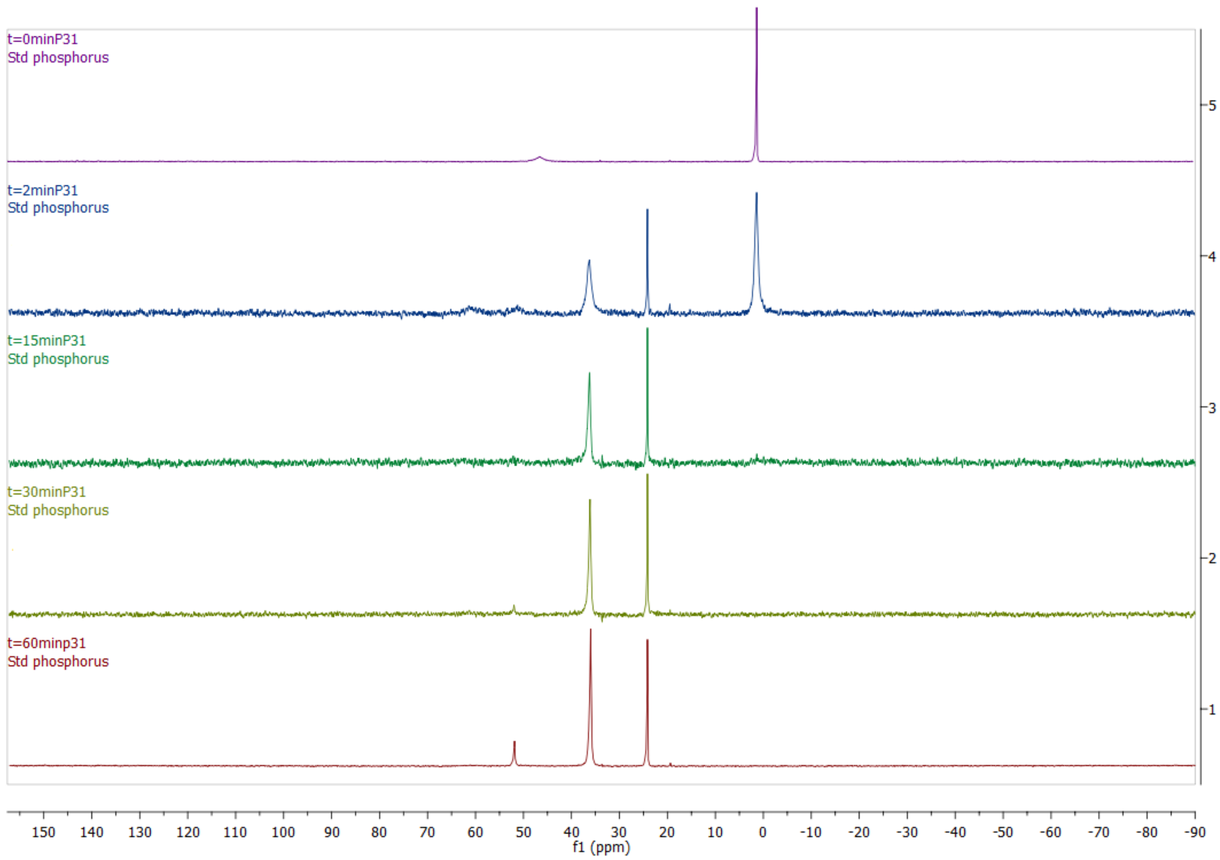

Figure S18. Representative time resolved ^{31}P NMR spectra for CO_2 hydrogenation with $Co(dmpe)_2H$. Catalytic conditions; 1.42×10^{-5} mol catalyst, 710 mM $P_1{}^{t}Bu$, 300 μL THF-d_8, 20 atm CO_2:H_2 (1:1 ratio), 21 °C. Key species: $P_1{}^{t}Bu$, 2.2 ppm; $Co(dmpe)_2H_2{}^{+}$, 52/61 ppm (*d*); $Co(dmpe)_2CO$, 53 ppm; $Co(dmpe)_2H$, 48 ppm; $P_1{}^{t}BuH^{+}$, 37.1 ppm.; $OP_1{}^{t}Bu$, 25.0 ppm.

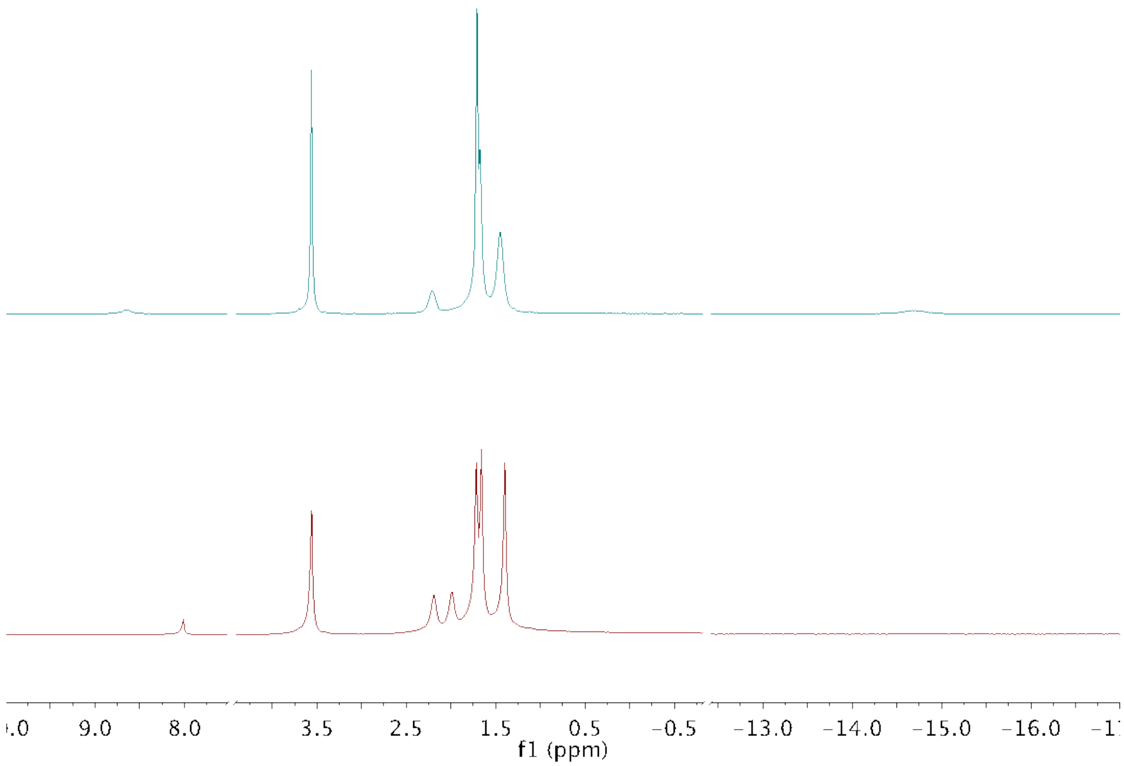

Figure S19. Proton NMR of the reaction of $Co(dmpe)_2H$ with H_2/CO_2 (1 atm) at -40 °C in THF-d_8 (lower spectrum) and warmed to 25 °C (upper spectrum). Key species; $HCOO^-$ bound to cobalt 8.17 ppm; $HCOO^-$, -8.8 ppm; $Co(dmpe)_2H_2^+$, -14.5 ppm.

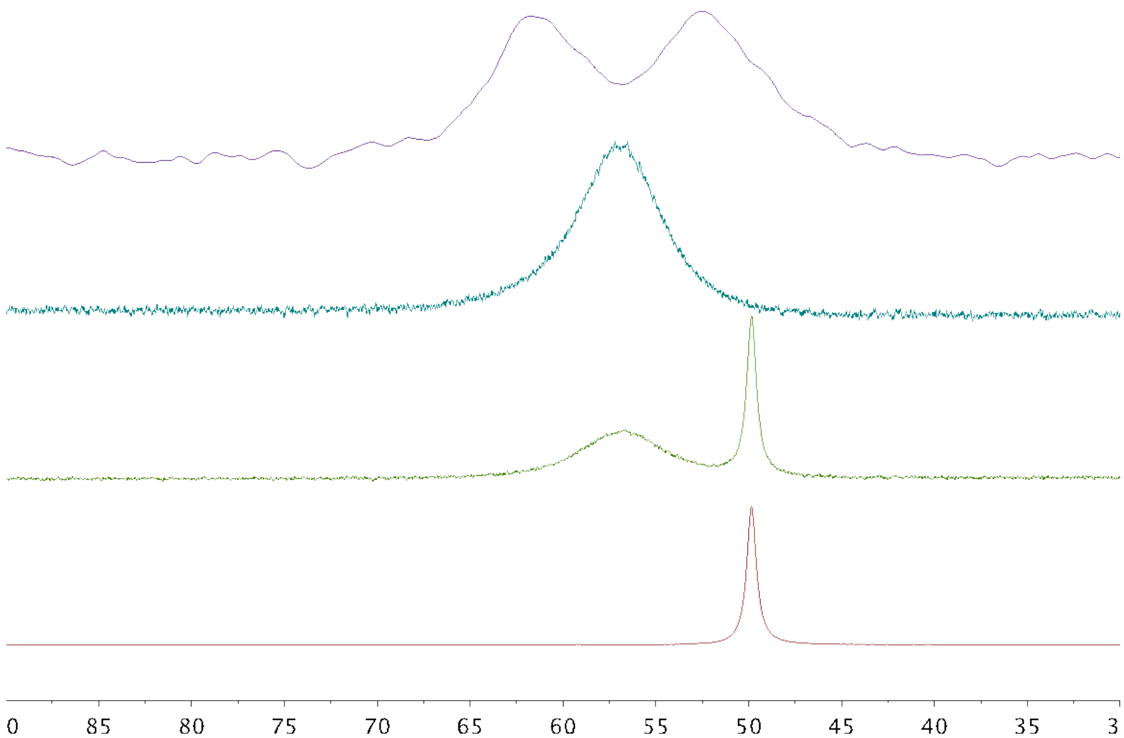

Figure S20. $^{31}P\{^1H\}$ NMR spectra for reaction of $Co(dmpe)_2H$ with H_2/CO_2 (1 atm) at -40 °C in THF-d_8. Lower spectrum is $Co(dmpe)_2H$ at -40 °C before addition of gas. Next spectrum is taken immediately after addition of gas with both the $Co(dmpe)_2H$ and the bound $Co(dmpe)_2(OOCH)$ present. The upper spectrum at 25 °C shows only $[Co(dmpe)_2H_2][HCOO]$.

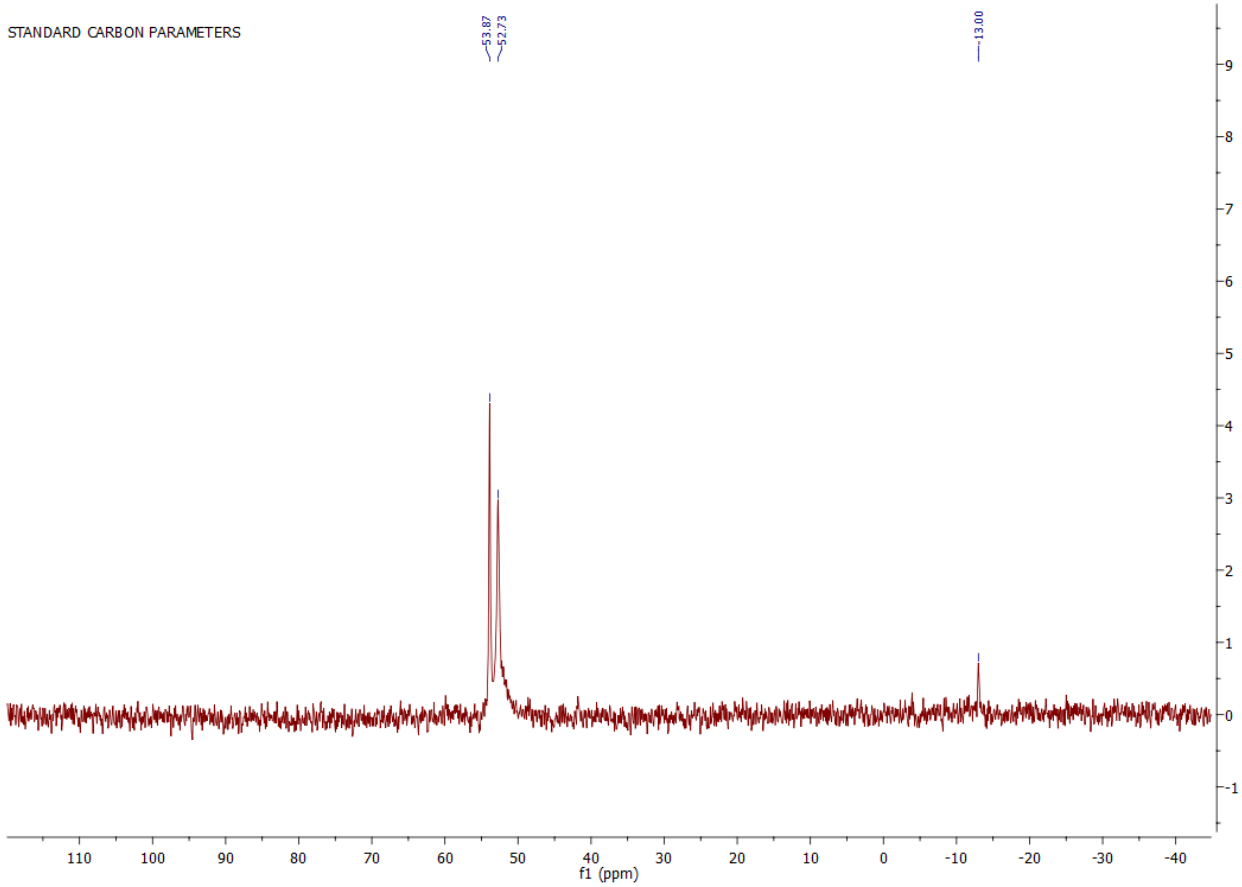

Figure S21. $^{31}P\{^1H\}$ NMR spectrum of $[(\mu\text{-dmpe})(Co(dmpe)_2)_2]^{2+}$ and $[Co(dmpe)_2CO]^+$ in CD_3CN (referenced to H_3PO_4 in D_2O).

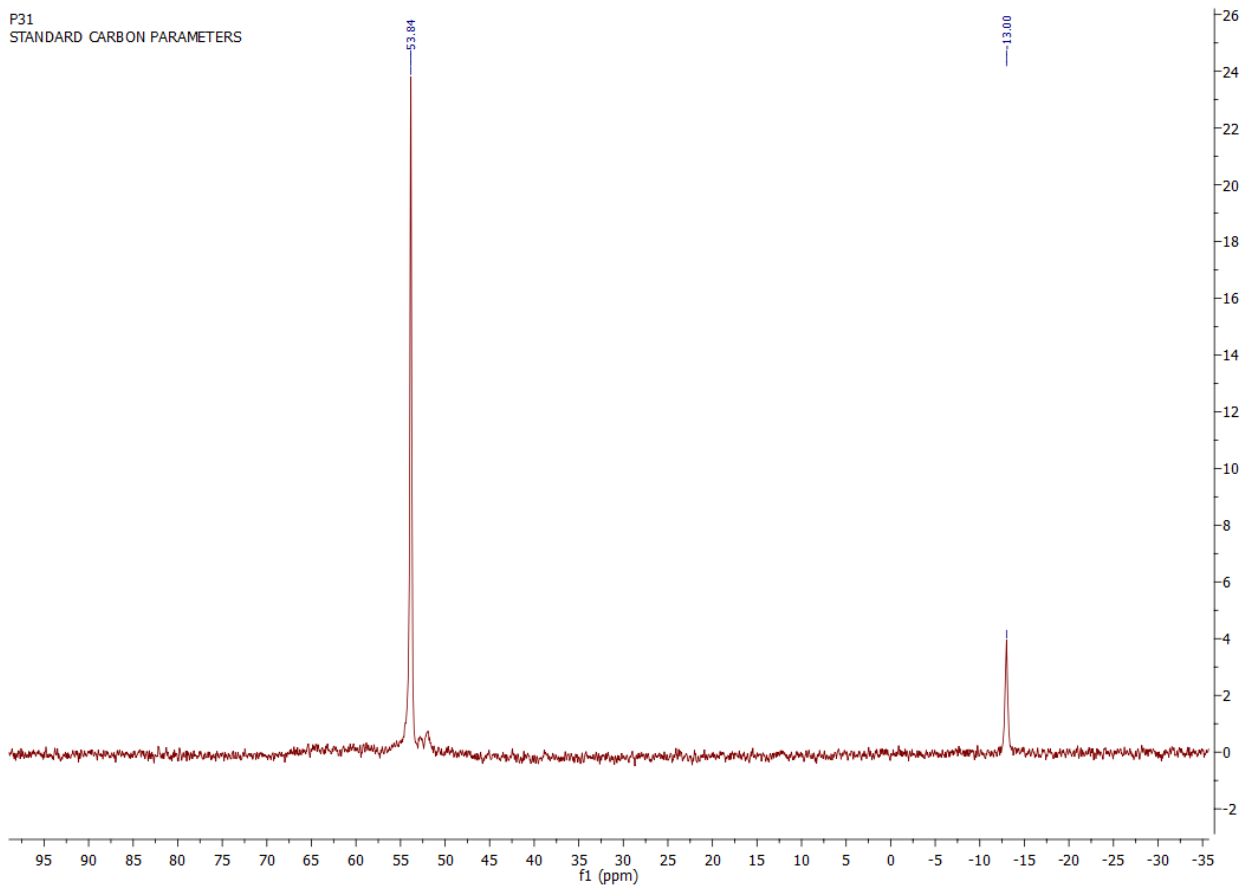

Figure S22. $^{31}P\{^1H\}$ NMR spectrum of isolated crystals from the mixture in fig S21 in CD_3CN (referenced to H_3PO_4 in D_2O).

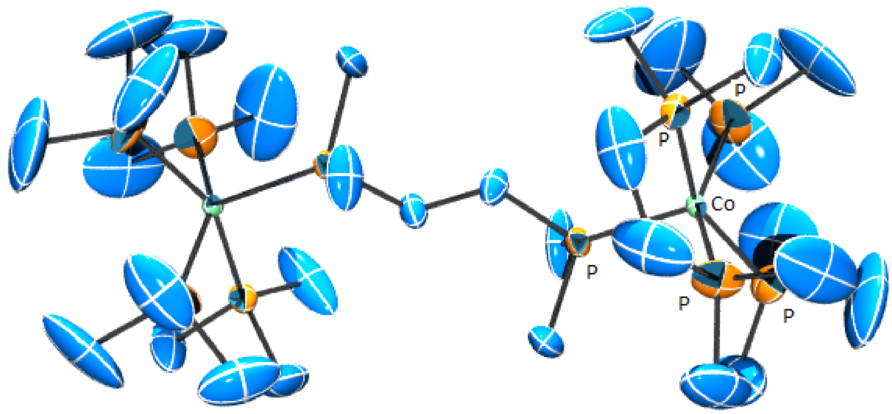

Figure S23. Crystal structure of [(μ-dmpe)(Co(dmpe)$_2$)$_2$]$^{2+}$ displaying connectivity. Single crystals of [VkdH]+ [HCO2] were grown by slow evaporation of diethyl ether into acetonitrile. A suitable crystal was selected and mounted on a nylon fiber using Paratone N oil on a Bruker APEX-II CCD diffractometer. The crystal was kept at 140.0 K during data collection. Using Olex2 [1], the structure was solved with the olex2.solve [2] structure solution program using Charge Flipping and refined with the ShelXL [3] refinement package using Least Squares minimization. Crystal Data for C15H40BCoF4P5 (M =521.06): monoclinic, space group C2/n (no. 15), a = 16.3392(14) Å, b = 21.3483(18) Å, c = 15.3507(12) Å, β = 116.251(4)°, V = 4802.3(7) Å3, Z = 8, T = 140.0 K, μ(MoKα) =1.078 mm-1, Dcalc = 1.441 g/mm3, 36242 reflections measured (3.372 ≤ 2Θ ≤ 54.482), 5350 unique (Rint =0.0691, Rsigma = 0.0472) which were used in all calculations. The final R1 was 0.1248 (I > 2σ(I)) and wR2 was 0.3304 (all data).1. Dolomanov, O.V., Bourhis, L.J., Gildea, R.J, Howard, J.A.K. & Puschmann, H. (2009), J. Appl. Cryst.42, 339-341. 2. Bourhis, L.J., Dolomanov, O.V., Gildea, R.J., Howard, J.A.K., Puschmann, H. (2013). in preparation 3. Sheldrick, G.M. (2008). Acta Cryst. A64, 112-122

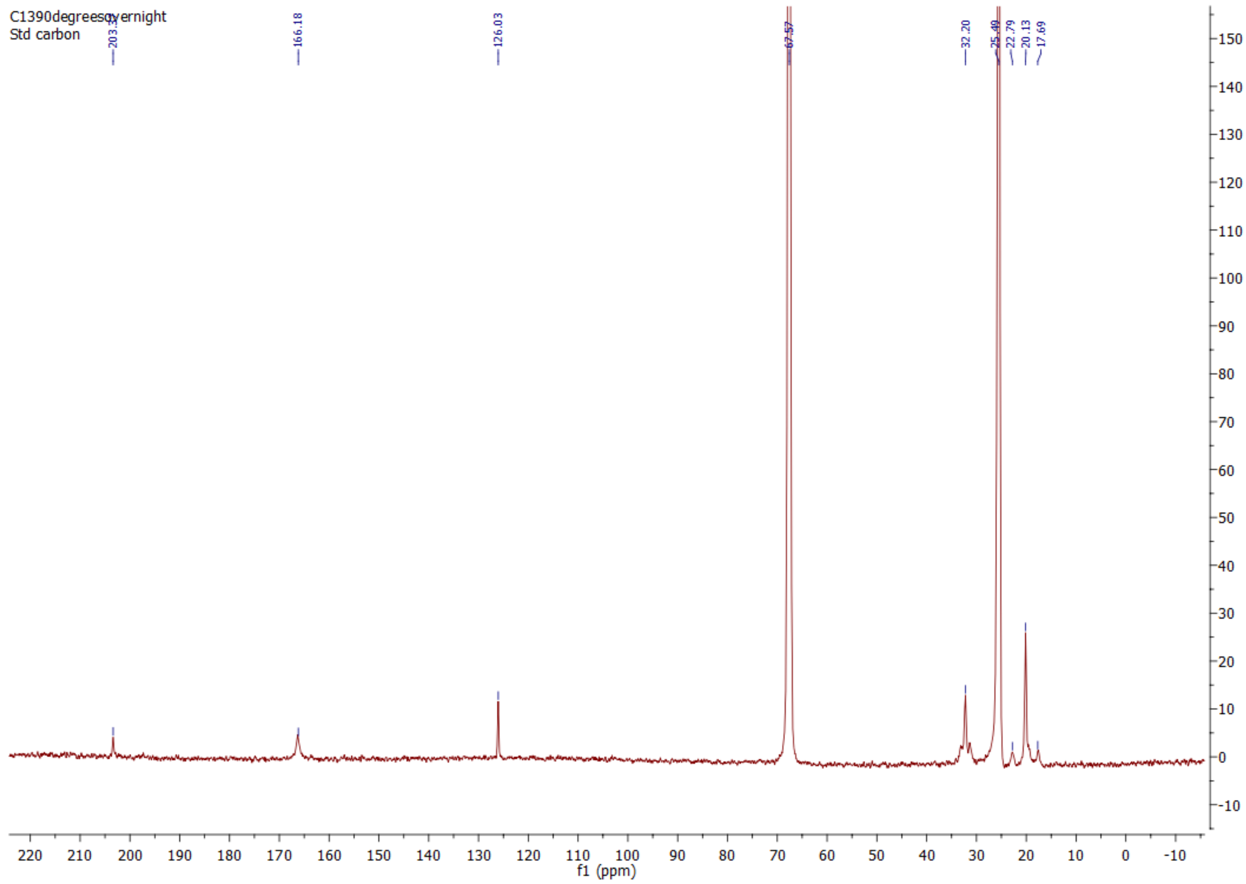

Figure S24. $^{13}C\ \{^1H\}$ NMR spectrum of $Co(dmpe)_2CO^+$ in THF-d_8 (10 s delay time). $Co(dmpe)_2CO^+$ was generated by adding 1.8 atm of 1:1 CO_2:H_2 to $Co(dmpe)_2H$ and letting the mix sit for 60 days. Key species: Co-CO, 203 ppm; $HCOO^-$, 166.3 ppm; CO_2, 126 ppm.

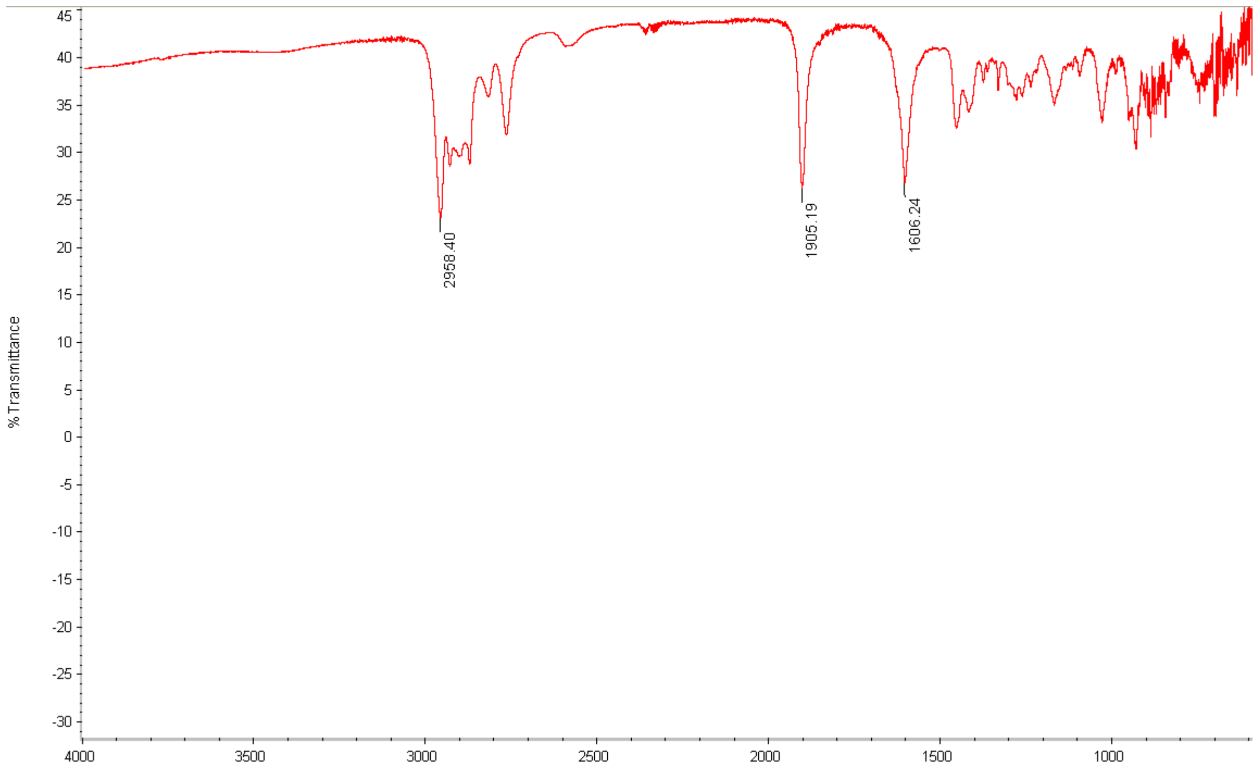

Figure S25. IR spectrum for $Co(dmpe)_2CO^+$, KBr Pellet.

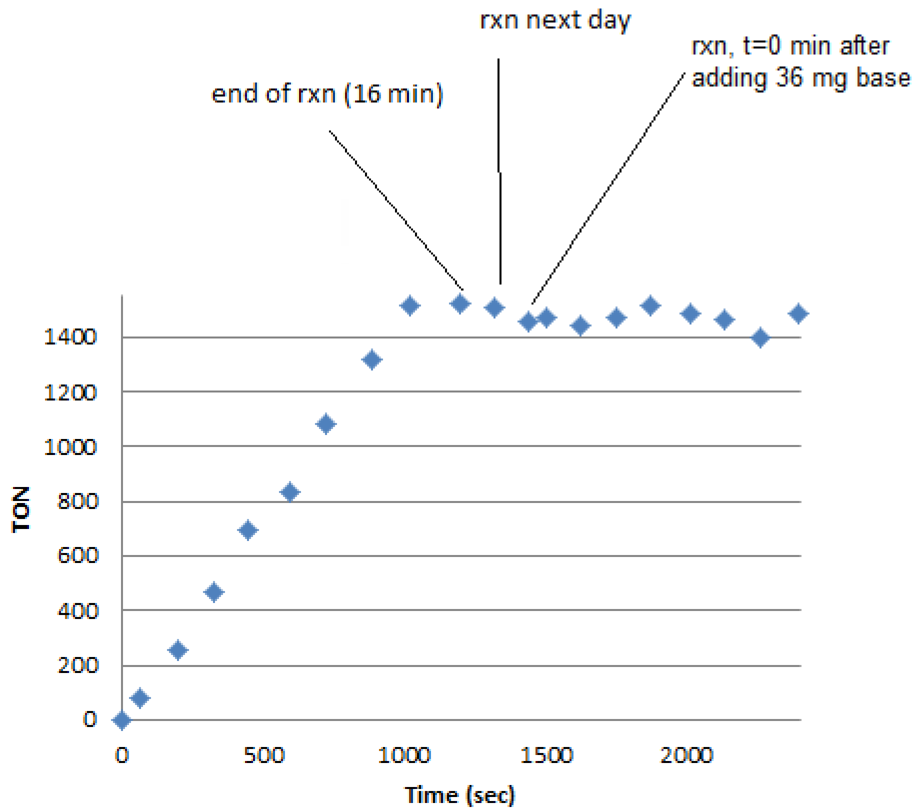

Figure S26. CO_2 hydrogenation kinetics using Co(dmpe)$_2$H as a function of TON. Catalytic conditions; 0.28 mM catalyst, 390 mM Vkd, 500 μL THF-d_8, 1.8 atm CO_2:H_2 (1:1 ratio), 21 °C. The reaction was complete after 16 min and then allowed to sit. After 24 h, the gas was removed and an additional 36 mg of base was added. Then the gas mix was reintroduced.

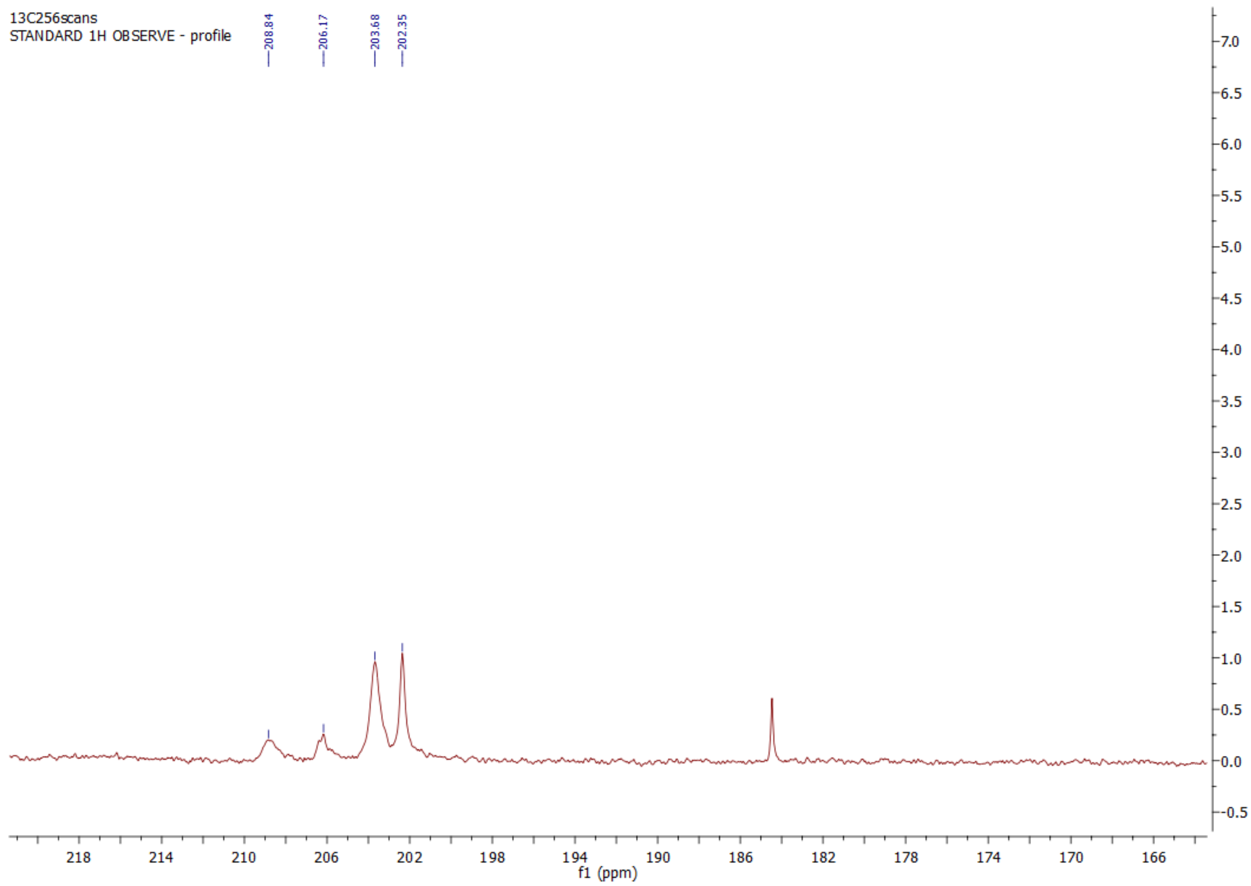

Figure S27. ^{13}C {^{1}H} NMR spectrum of $[Co(dmpe)_2CO]^+$ (and other species) in THF-d_8 after adding 1 atm of ^{13}CO to $[Co(dmpe)_2][BF_4]$. Close up of the C=O region displays over substitution of CO for dmpe occurs resulting in multiple Co-CO species.

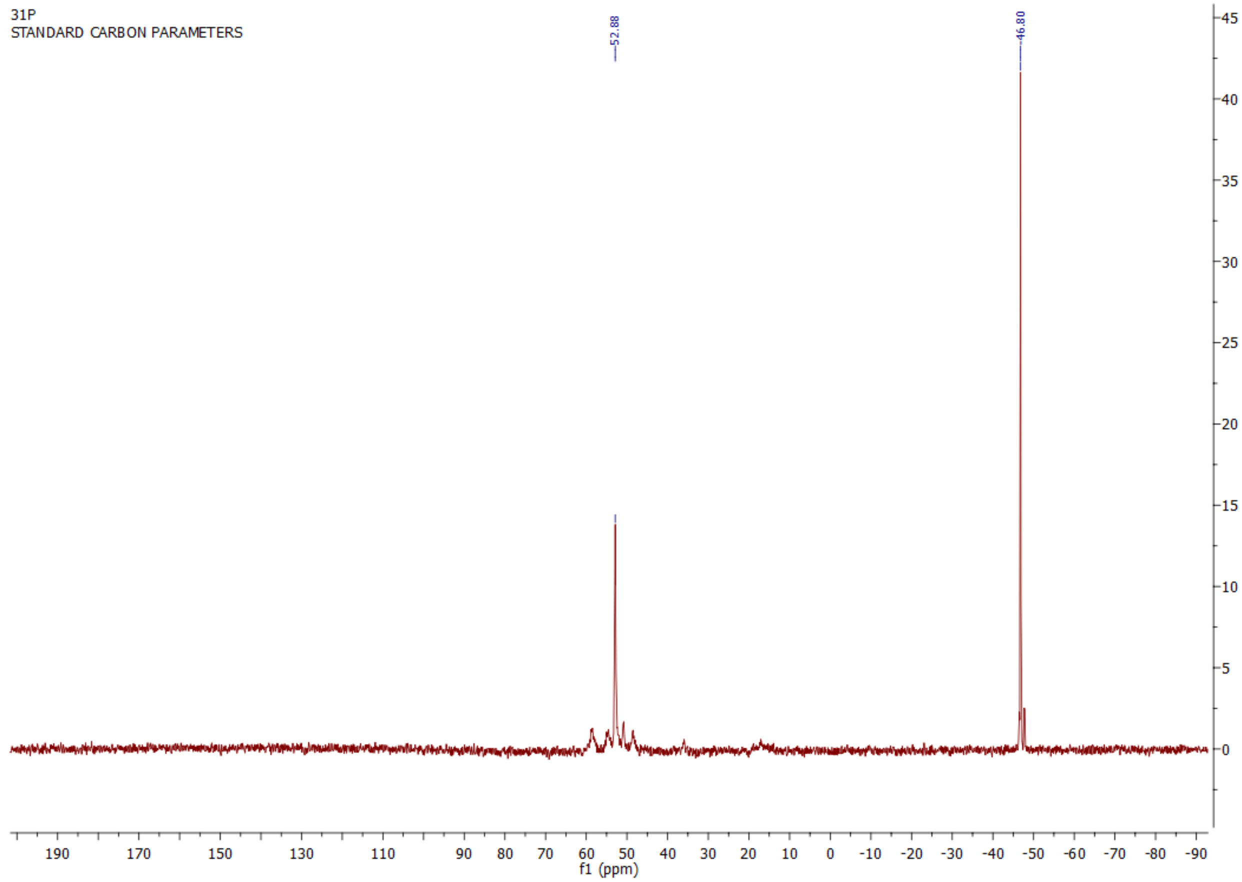

Figure S28. $^{31}P\{^{1}H\}$ NMR spectrum of $[Co(dmpe)_2CO]^{+}$ in THF-d_8 after adding 1 atm of ^{13}CO to $[Co(dmpe)_2][BF_4]$. Free dmpe at -46 pm is a result of over substitution of CO for dmpe.

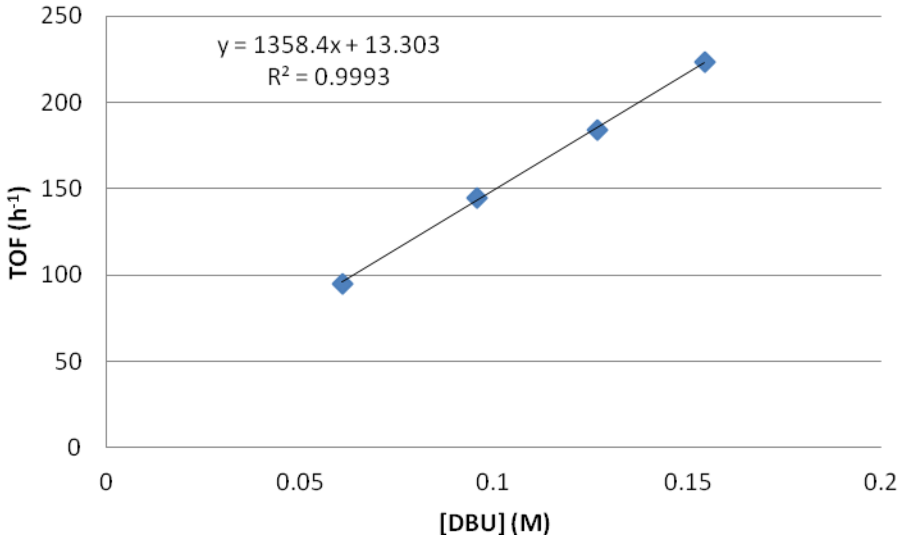

Figure S29. First order dependance on DBU concentration (as a function of TOF).

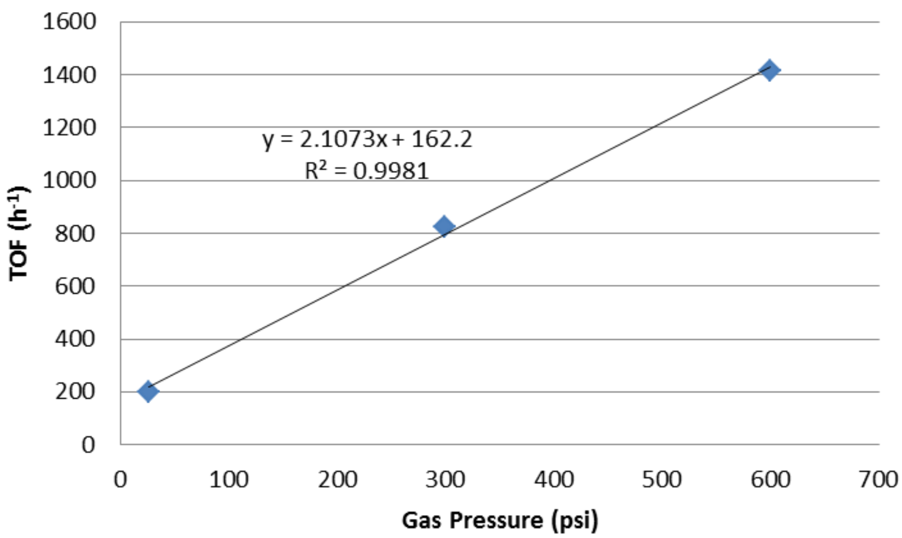

Figure S30. First order dependance of gas pressure as a function of TOF (with $P_1{}^t$Bu).

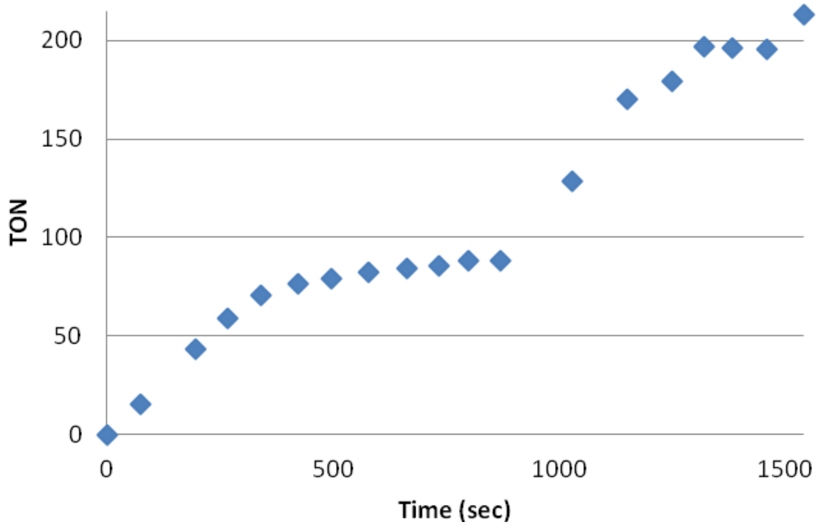

Figure S31. CO_2 hydrogenation kinetics using Co(dmpe)$_2$H as a function of TON. Catalytic conditions; 1.39×10^{-6} mol catalyst, 559 mM Vkd, 500 μL THF-d_8, 1.8 atm CO_2:H_2 (1:1 ratio), 21 °C. The tube was only refilled once at 1029 sec. The purpose was to determine how long it takes to consume all the gas and to validate our method.